Extended Methods and Preserved Exposition

Companion to the 12 October revision. This file retains the complete pre-condensation methodology, discussion and chronology. The current manuscript governs conclusions. No measurements or original numerical requirements were changed by condensation. Diagrams and tables remain in the manuscript; their captions are retained here for orientation.

Chapter 3—Methodology

3.1 Research Design

The research used a staged-freeze design to evaluate three aspects of one artifact. RQ1 compared representations at validation-selected thresholds; RQ2 assessed distribution monitoring and future-only routing; RQ3 measured the real Hypertext Transfer Protocol (HTTP) path. This chapter specifies the information flow, source preparation, models and estimands, followed by the accepted measurement structure and the separate follow-up comparisons. The original matrix combines source results and operational cells 1–72 from the fourth amended attempt with cells 73–125 from the October 1 continuation. Models, thresholds, workloads and decision rules remained fixed across these two sessions. Acceptance applied to the complete eligible prefix, not to favorable outcomes. Section 4.6 records the methodological changes and session structure; the history supplement retains the underlying preparation and execution records.

The unit of analysis for detection is a retained uniform resource locator (URL) record. The clustering unit for primary recall intervals is the registrable domain. The unit for monitoring is a 256-request window with stride 64. The unit for the H3 operational error rate is one measured HTTP request under the frozen request construct.

3.1.1 Architecture and Information Flow

The architecture separates four functions: preparation, inference, monitoring and evaluation. Preparation determines record eligibility and preserves source identities. Inference maps an eligible URL to a score and an allow/alert decision. Monitoring observes the feature distribution and may alter future routing, but does not change reference labels or earlier decisions. Evaluation joins saved decisions to the declared outcome strata and separately measures service execution. This separation prevents model scores, routing actions and publisher-assigned labels from being treated as the same form of evidence.

Figure 3.1 shows the implemented dataflow. Each URL produces 25 structural features and a Logistic-L1 score. The fixed uncertainty rule either retains the first-stage decision or requests character-transformer inference. The Gaussian mixture model (GMM) observes the structural features and first-stage score in complete windows; an alert affects only the next 256 requests. Paired outputs support internal and external detection comparisons, while HTTP replay measures service execution. The distinction between logical routing and physical inference is retained in the recorded counters.

Figure 3.1. Frozen inference, future-only routing and separate evaluation boundaries.

Table 3.1. Reading map from technical component to experiment and evidence.

The decisive information-flow restriction is that evaluation strata do not select routing. After mechanical validity and conflict exclusions, every retained external record participates in the same ordered stream; gold, certified and other outcome groups are applied to its saved outputs afterward. Thus a reported certified-negative error rate answers how the common policy treated that stratum, rather than how a separate source-aware policy performed. The distinction is especially important here because source composition and operational behavior differ sharply from the internal evaluation.

3.2 Outcome-Label Contract

The local binary field is_phishing records the source's reference designation. A local value of 1 means phishing and a local value of 0 means legitimate. For PhiUSIIL, Native label 0 maps to local is_phishing=1. Native label 1 maps to local is_phishing=0. The transformation is a deterministic lookup identified as phiusiil-native-label-map-v1. Neither the investigator nor another individual assigns, overrides, or changes a retained confirmatory outcome.

These labels are publisher-provided, source-derived reference classifications rather than independent forensic judgments. They define reproducible evaluation outcomes under the source contract. Unknown, missing or unmapped labels are quarantined and reported mechanically, never inferred from URL appearance.

The external preparation protocol requires the PhishVN schema and categorical values to be verified before mapping. National Cyber Security Center (NCSC) gold phishing designations map to is_phishing=1, and certified trusted-registry designations map to is_phishing=0 for primary external inference. Other tiers retain only their frozen

secondary roles. Tranco remains a separate reference-negative control and is not promoted to verified benign ground truth.

An external record must have a valid URL, source, source class, confidence tier, published split, and stable published record ID. The verified source-class and confidence-tier pair must be one of the combinations allowed by the frozen mapping: NCSC gold and certified trusted-registry records for the primary strata; NCSC silver and ChongLuaDao or OpenPhish bronze records for secondary or sensitivity analysis; and Tranco only as the separately designated reference-negative control. A missing, unknown, or undefined combination is quarantined rather than interpreted.

The common conflict rules operate at group level. A canonical URL with conflicting normalized mappings, including conflicting source classifications when those fields are present, is quarantined as a whole. A registrable-domain group assigned to more than one development split or more than one published external split is also quarantined as a whole. The same rule applies when an external registrable domain overlaps a PhiUSIIL registrable domain. For exact duplicates with the same mapping, the retained record is the one with the lexicographically smallest stable identifier, whether that identifier was published or deterministically derived. If no stable identifier exists, the entire duplicate group is quarantined. No numeric label encoding is assumed for the external source. No personal exception or discretionary override is permitted.

These schema, validity, duplicate, and conflict rules define quarantine before routing. Once the retained stream is fixed, source and confidence designations are used after

label-blind routing only to form analytic outcome strata; they do not remove retained records from routing.

3.3 Development Source and Preservation

PhiUSIIL, UCI Machine Learning Repository dataset 967, is the sole source used for fitting, validation and the internal held-out evaluation (Prasad & Chandra, 2024b). It is distributed under CC BY 4.0 and is associated with DOI 10.1016/j.cose.2023.103545. The retained archive and embedded CSV have separate SHA-256 identities, supplied in Appendix C so that the source can be authenticated without interrupting the methodological argument.

The publisher reports Open PageRank as the legitimate source and PhishTank, OpenPhish, and MalwareWorld as phishing sources. It reports phishing retrieval from October 1, 2022, through May 21, 2023 (Prasad & Chandra, 2024a). No corresponding legitimate collection window is established in the public record used here. Public per-row source, timestamp, snapshot, and independent-adjudication fields are not available. The preserved source, DOI, source description, mechanical mapping, and this limitation together define the development-data provenance statement.

3.4 Canonicalization and Quarantine

Preparation accepts absolute HTTP or Hypertext Transfer Protocol Secure (HTTPS) URLs. It lowercases the scheme, normalizes the host using Internationalizing Domain Names in Applications (IDNA), normalizes numeric ports, removes default ports, represents an empty path as a slash, and uppercases percent-escape hex without decoding it. User information, path spelling, query order, and fragments are preserved. A canonical

key is used only for duplicate handling and group consistency; it is not passed to the detector as a substitute label.

An invalid or unsupported URL is quarantined. Records that reduce to the same canonical key and the same mapped label are treated as a same-label duplicate group, from which one record is retained according to the frozen rule and the remainder are quarantined. If one canonical group contains conflicting mapped labels, the entire group is quarantined. Every exclusion receives a mechanical reason. No visual review or personal judgment resolves conflicts.

The original external preparation applied these rules to the publisher's raw url cells and reached the September 28 whole-study hold. The separately authorized external amendment uses the publisher's exact url_norm for both parsing and model inputs, retaining the original raw cells as provenance. It permits no local scheme repair, trimming or raw-value fallback and leaves mapping, overlap and quarantine rules otherwise unchanged. Section 4.6 records the preparation counts that informed the amendment and the later execution history.

3.5 Registrable-Domain Allocation

Registrable domains are computed with the pinned Public Suffix List identified in Appendix C. Retained domain groups are ranked by SHA256 of the fixed allocation identifier, the seed 20260816, and the domain. Hamilton largest-remainder apportionment converts target shares of 70%, 15%, and 15% to integer counts for train, validation, and group test. Ties are ordered train, validation, group test. Allocation is label-blind and is not stratified, rerolled or rebalanced.

The train partition may fit preprocessing and models. The validation partition may select thresholds, early stopping, the cascade band, and the GMM alert threshold under its separate frozen development contract. The group-test partition remains excluded from all fitting and selection.

Two broad text searches exposed rows from the ignored local group-test file. The first occurred on September 3, 2026, after the baseline contract and implementation were frozen. The second occurred during a local wording search on September 9, 2026. The partition had not yet been model-scored, but these displays established analyst exposure; it could no longer be described as unseen. The displayed rows prompted no change to a model, threshold, gate, routing rule or scientific procedure. The separate v2 transformer publication correction arose from code review and changed no scientific field. Neither search performed fitting, scoring, metric computation or PhishVN access.

The original protocol called for one noninteractive processing pass after all four RQ1 models, thresholds, evaluator, manifest specifications, selection rules, environment and hashes were frozen. Interrupted study attempts later repeated source scoring under unchanged scientific rules, as Section 4.6 records. The partition is now both analyst-exposed and model-scored. The realized study cannot be described as a single lifetime pass over an unseen test set.

3.6 Detection Models

The structural extractor returns an ordered 25-element float64 vector. Each value has one of three stated bases: the supplied URL string, urlsplit component text, or the existing IDNA-normalized ASCII hostname. It neither fetches a destination nor substitutes a

canonical URL for the supplied string. The accepted baseline contract fixes feature order and input interpretation; its earlier versions and exact identities are recorded in Appendix C.

The feature order is fixed. Positions 1 through 6 measure string and component lengths: raw_url_codepoint_length, raw_url_utf8_byte_length, hostname_ascii_length, path_codepoint_length, query_codepoint_length, and fragment_codepoint_length.

Positions 7 through 12 count hostname, path and query elements: hostname_label_count, hostname_ascii_digit_count, hostname_hyphen_count, hostname_punycode_label_count, path_segment_count, and query_parameter_count.

Positions 13 through 20 describe character composition and escaping: raw_url_ascii_letter_count, raw_url_ascii_digit_count, raw_url_other_codepoint_count, raw_url_ascii_digit_ratio, raw_url_other_codepoint_ratio, raw_url_unique_codepoint_count, raw_url_utf8_byte_entropy_bits, and percent_escape_count.

Positions 21 through 25 indicate scheme and component presence: is_https, has_userinfo, has_explicit_port, has_query_delimiter, and has_fragment_delimiter.

These features provide a reproducible local structural baseline, not an optimal feature set. Several are dependent, so individual coefficients are not causal feature importance. Reusing the accepted Logistic-L1 model fixes the first-stage comparison; reconstruction must preserve its authoritative scoring procedure because even mathematically equivalent formulas can alter floating-point ties.

The contract accepts only the local raw_url field under the existing canonical-url-v1 preparation rules. For amended external evaluation, that local input contains the publisher's exact url_norm; the original publisher url cell remains separate provenance, as described in Section 3.4. A missing or invalid value raises the fixed error, "raw_url is missing or invalid under canonical-url-v1." Values are not imputed. The forbidden predictors comprise label and outcome fields; split membership; record, canonical-URL, and registrable-domain identity; and all source, source-class, confidence, published-split, publisher, and publisher-label fields. These exclusions prevent the baseline from learning the prepared label or collection metadata instead of URL structure.

Length-only and Logistic-L1 use the same fitted pipeline. StandardScaler(with_mean=True, with_std=True) is followed by LogisticRegression(solver="saga", penalty="l1", C=1.0, class_weight="balanced", fit_intercept=True, max_iter=5000, tol=1e-4, random_state=42). Length-only uses only raw_url_codepoint_length; Logistic-L1 uses all 25 features in contract order. The variance-reduced stochastic solver SAGA uses an unpenalized intercept. A convergence warning is an error, every fit must stop before max_iter, and fitted state must be finite. There is no hyperparameter or feature search. Training data alone fit the scaler and classifier; validation data alone select the threshold.

The accepted SAGA specification followed a nonconvergent liblinear fit and a training-only scoring diagnostic. The configuration was fixed before the accepted validation run; the earlier failures did not enter the reported generalization estimates. Appendix C retains the solver, warning and chronology details. This development history is distinct from evidence that the resulting model transfers to another population.

The scoring-integrity policy rq1-scoring-integrity-v1 permits only three exact RuntimeWarning messages emitted during decision_function or predict_proba on macOS arm64 with Accelerate. The warnings are recorded, the returned values must be finite, and scikit-learn decision values and the full two-column probability matrix must agree with independent float64 calculations at rtol=1e-12 and atol=1e-12. Warnings during scaling, fitting, threshold selection, or any other stage remain fatal.

The accepted character-model and cascade specification is rq1-transformer-cascade-v2. Its scientific model, selection and artifact rules are stated here; Appendix C records the contract versions and publication chronology.

Stage 2 implements rq1-transformer-cascade-v2. Each raw URL is normalized under canonical-url-v1; remaining characters outside the American Standard Code for Information Interchange (ASCII) are percent-encoded as Unicode Transformation Format—8-bit (UTF-8) bytes while existing percent escapes and Request for Comments (RFC) 3986 reserved characters are preserved (Berners-Lee et al., 2005). The resulting ASCII sequence is limited to 256 characters by retaining the first 192 and last 64. The vocabulary is derived from training data only and ordered by ascending ASCII code point. It reserves PAD=0, UNK=1, and character IDs beginning at 2 and adds no other special token. Sequences are right-padded, and PAD positions are masked.

The model runs in PyTorch 2.7.1 in float32. It uses learned token embeddings of width 192 with padding_idx=0 and learned position embeddings of width 192 for absolute positions 0 through 255. Four pre-normalization encoder layers each use six attention heads, feed-forward width 768, Gaussian error linear unit (GELU) activation, and

dropout 0.1. A final LayerNorm precedes masked mean pooling over non-PAD positions and a linear-logit head. Embeddings are initialized from a normal distribution with mean 0 and standard deviation 0.02. Linear and attention weights use Xavier-uniform initialization, biases are zero, LayerNorm weights are one with zero bias, and the PAD embedding row remains zero through padding_idx.

The official transformer runtime uses Apple's Metal Performance Shaders (MPS) backend. Automatic mixed precision is disabled, the data loader uses zero data-loader workers, deterministic algorithms are enabled with warn_only=False, and the primary seed is 42. Training minimizes binary cross-entropy with logits using pos_weight=train_negative_count/train_positive_count. Optimization uses AdamW's decoupled weight-decay formulation (Loshchilov & Hutter, 2019). One AdamW parameter group contains every trainable parameter with learning rate 1e-4, betas (0.9, 0.999), epsilon 1e-8, weight decay 0.01, and no scheduler. Training batch size is 256, validation batch size 512, training runs for a maximum of 40 epochs, and clipping limits the gradient norm at 1.0. One generator is seeded 42 once before epoch 1 and advances across epochs; training shuffles, validation remains in pinned source order, and neither loader drops its final batch.

Validation average precision is measured after every epoch with dropout disabled. Epoch 1 establishes the recorded best and checkpoint. A later epoch qualifies only when its average precision is strictly greater than the recorded best plus min_delta=1e-4. A qualifying epoch resets patience to zero; each consecutive nonqualifying epoch increments it, and training stops immediately after the fifth consecutive nonqualifying epoch. The earliest qualifying best is preserved and restored. Any warning during

transformer training or validation, nonfinite value, missing class, or deterministic-algorithm error stops the run without publication. The separately audited stage-one scoring exceptions remain limited to the policy stated above.

Only the pinned training and validation partitions, preparation summary, structural model and model contracts enter character-model development. Group-test and external records are excluded from this fitting interface. Stage 1 is reconstructed from the accepted Logistic-L1 artifact and is never refitted during character-model development.

Artifact validation binds the fitted state, preprocessing, threshold and reported predictions to the same accepted model. A partial or failed output does not become a reported fit. The durable-publication implementation and its execution history are retained in Appendix C.

The evidence distinguishes completed model fitting from installation of its public summary. Failed predecessors remain separate from accepted model artifacts; publication status is not a predictive-performance result.

The private output directory has mode 0700. Its vocabulary.json, transformer-weights.npz, transformer.json, cascade.json, and SHA256SUMS files have mode 0600. JavaScript Object Notation (JSON) is canonical UTF-8 with sorted keys, compact separators, a terminal newline, and nonfinite values rejected. The weight archive stores the complete state dictionary as lexicographically ordered, C-contiguous little-endian float32 tensors in uncompressed NPY 1.0 members with fixed metadata. SHA256SUMS lists the other four files in lexical order and does not include itself. The public summary may contain only counts, rates, configuration, versions, hashes, warnings, and status. It excludes URLs, records, domains, sequences, predictions, coefficients, and weights.

The fitting implementation and verification record are versioned with the research package. Appendix C records their commit identities and the controlled scoring correction. Code validation establishes implementation consistency, not external detection efficacy.

3.7 Threshold and Cascade Calibration

For each detector, the model-assigned probability score is P(is_phishing=1), and score >= threshold produces an alert. This notation does not imply that the score is calibrated to deployment risk. Candidate thresholds are the unique validation scores plus the finite no-alert value nextafter(maximum validation score, +infinity). Selection maximizes validation recall subject to the exact one-sided 95% Clopper-Pearson false-positive-rate (FPR) upper bound being <= 1%. Ties use the smaller FPR upper bound and then the higher threshold. If no candidate qualifies, the recorded outcome is target_not_met; the rule is not relaxed.

The cascade is calibrated only if both the stage-1 and transformer thresholds have status selected. If either threshold is target_not_met, no cascade is calibrated or accepted and the cascade also records target_not_met. Otherwise, it considers symmetric uncertainty bands around the locked Logistic-L1 threshold. A record is inside the band when abs(stage1_probability - stage1_threshold) <= half_width; the boundary is inclusive. Inside the band, both probability and decision come from the transformer. Outside it, both come from Logistic-L1. Candidate half-widths are the sorted unique absolute distances between the Logistic-L1 validation probabilities and its threshold.

Among candidates satisfying cascade_recall >= transformer_recall - 0.02 and an exact one-sided 95% Clopper-Pearson FPR upper bound <= 1%, selection first minimizes transformer invocations and then chooses the smaller half-width. If no candidate qualifies, the cascade records target_not_met and is not accepted. Once selected, the band is applied unchanged to internal test, external, and replay records.

3.8 RQ1 Analysis

The September 21 inference-method amendment adopted the previously specified singleton OpenBLAS/MPS convention for subsequent no-fit evaluation and serving. The development comparison did not establish exact equivalence with the original batch-scoring convention: one transformer decision differed. Adoption was therefore a development-informed methodological change before protected evaluation, not a passed compatibility test. Historically selected weights, thresholds and the cascade band were carried forward without recalibration. Their optimality or checkpoint-selection equivalence under singleton scoring is not claimed. The accepted paired predictions use the common convention; historical development summaries retain their original meaning. The amendment did not authorize an alternative-runtime search or additional compatibility execution.

Length-only, Logistic-L1, transformer-only, and cascade predictions are paired by retained record. The original RQ1 plan specified one noninteractive group-test pass and one external pass after their respective freezes. Source scoring subsequently occurred in repeated interrupted study attempts; the first, second and fourth amended attempts accepted both source outputs. Each attempted source evaluation used the same frozen models and thresholds, with external source and confidence filters applied after routing to

form the outcome strata. Final synthesis uses the authenticated evidence selected under the disclosed recovery protocol, without choosing among attempts by their outcomes. Repeated execution does not restore test-set blindness or establish compliance with the original single-pass intention.

Bootstrap resampling follows the general framework of Efron (1979), with the domain-clustered estimand specified below. The original primary recall analysis used a registrable-domain-clustered percentile bootstrap using 2,000 replicates and seed 20260816. H1 uses the lower confidence bounds for the four prespecified differences and the observed FPR gates in Section 1.4.1. Average precision (AP), precision, F2, Matthews correlation coefficient (MCC), balanced accuracy, receiver operating characteristic area under the curve (ROC-AUC), Brier score, and calibration error are secondary descriptions.

The paired-recall implementation operates on saved binary predictions for the declared positive stratum. Record identifiers must be unique and match in the same order across both prediction vectors. Let D be the number of registrable domains represented in that stratum. Domains are ordered by canonical ASCII spelling, and each replicate samples D domains uniformly with replacement. Every occurrence of a selected domain contributes all its eligible positive rows. The replicate statistic is the sum of candidate-minus-reference decisions divided by the sampled row count. This preserves URL weighting when domains have unequal sizes; averaging domain-level recall would estimate a different quantity.

Each contrast uses a fresh NumPy 2.2.6 Generator(PCG64(20260816)), with one integers(0, D, size=D, dtype=int64) call for each of 2,000 replicates. Contrasts on the same positive stratum therefore use the same domain draws. The 2.5th and 97.5th percentiles use linear interpolation. An empty stratum has no estimate or interval. A stratum containing one domain retains its point estimate but has no estimable clustered interval. With at least two domains, the historical implementation returned any degenerate interval as calculated. The corrected decision analysis below does not use such an interval as evidence of population certainty or noninferiority. The procedure is implemented and checked on synthetic known-answer fixtures in paired_evaluation.py. Runtime, manifest, operational and secondary-analysis specifications were subsequently bound to the execution profile used for the attempts in Section 4.6; Chapter 4 reports the completed, verified evidence under the disclosed recovery boundary.

Post-review paired-inference correction. Identical observed predictions make every bootstrap replicate identical; resampling cannot introduce an error pattern absent from the sample. A zero-width interval therefore cannot establish the two-percentage-point population recall margin. The original outputs remain in the research record, but the corrected adjudication replaces that interpretation. The original research questions, outcome strata, numerical margins, observed-rate gates and two-sided 95% confidence convention are retained. This is an analysis correction after review, not a claim that the replacement method was prespecified before the study.

For nonstateful gold contrasts, let b denote candidate-only detections and c reference-only detections among n labeled positive pairs. The estimand is the difference between their detection probabilities, estimated by (b-c)/n. Each discordant count has a binomial

marginal when pairs are independent. Clopper–Pearson bounds allocate 0.05/4 to each of the four marginal tails; projecting their simultaneous rectangle gives [L(b)-U(c), U(b)-L(c)]. The union bound supplies at least 95% coverage under independent, identically distributed paired sampling, without assuming independence between b and c. This conservative envelope is not an optimized shortest exact interval. Newcombe's (1998) method 10 is reported separately as an approximate-score sensitivity analysis, not as the source of the exact guarantee. For 69 concordant positive pairs, the selected envelope is [-6.1533, 6.1533] percentage points; method 10 gives [-5.2737, 5.2737]. Neither establishes the -2-point lower-bound requirement. Failure to establish that margin does not demonstrate that the cascade is inferior.

The 69 gold rows belong to distinct registrable domains, but domain uniqueness alone does not establish campaign independence or label accuracy. Those are assumptions of the gold-pair intervals. The nondegenerate internal structural-versus-length contrast retains the original whole-domain bootstrap and URL-weighted estimand. An independent-row formula is not substituted for repeated-domain internal data or for a policy whose predictions depend on preceding stream windows. Their degenerate zero contrasts retain their observed estimates without an established population lower bound. Likewise, historical row-wise binomial FPR bounds are not population-coverage guarantees for the 2,497 certified negatives in 234 domains. Original FPR decisions continue to use observed rates, as specified. These are contrast-specific intervals, not simultaneous 95% coverage for all 22 checks; the joint hypotheses still require every constituent condition.

3.9 GMM Monitoring and Future-Only Routing

The GMM development method is frozen in rq2-gmm-development-v1 under protocol v1.10. Its 26 inputs are the 25 structural features in Section 3.6, in contract order, followed by the frozen Logistic-L1 phishing probability. Stage 1 is reconstructed from its accepted artifact and is not refitted. The GMM's portable stage-one probability path is unchanged; the separate RQ1 scoring diagnosis does not rescore or revise the GMM result. A separate StandardScaler fits the entire 26-column training matrix only, using population variance (ddof=0) and a scale of one for constant columns. Fitting and scoring use central processing unit (CPU) float64 with one numerical-library thread and pinned NumPy 2.2.6, SciPy 1.15.3, and scikit-learn 1.7.2. The contract pins the NumPy wheel and requires its scipy-openblas 0.3.29 backend; a different backend is rejected before research inputs are read.

Validation domains are allocated without labels, class counts, or rerolls. Let D be the number of unique normalized ASCII registrable domains. Each domain is keyed by SHA-256 of the UTF-8 namespace rq2-gmm-validation-v1, a NUL byte, the ASCII seed 20260816, another NUL byte, and the ASCII domain. Domains are sorted by digest bytes and then domain bytes. The first floor(D/2) domains form monitor calibration, and the remainder form the independent false-alert audit. Within each stream, rows retain their pinned validation source order.

Six diagonal-covariance Gaussian mixtures are fitted on training data only, with component counts from one through six. Every candidate uses five k-means initializations, random state 42, tolerance 0.001, covariance regularization 0.000001 and a maximum of 500 iterations. The candidate with the lowest training Bayesian

information criterion (BIC; Schwarz, 1978) is selected; an exact tie favors the smaller component count.

All candidates must converge with finite parameters, positive variances and weights, and normalized weights. If any candidate fails, the entire run stops without skipping that candidate or retrying. Warnings, overflow, invalid arithmetic, division errors and nonfinite values are fatal. Numerical underflow is ignored only during mixture fitting, BIC calculation and likelihood scoring, where negligible mixture contributions may become zero. It remains strict elsewhere.

Complete windows contain 256 requests with stride 64 and are scored by the float64 mean negative log-likelihood. Each stream must provide at least one complete window. The calibration boundary is the 95th percentile of calibration-window scores using NumPy's linear quantile method; a score strictly greater than the boundary is an alert. The independent audit passes the false-alert gate only if 20 * alert_windows <= complete_windows. Its overlapping windows are counted separately, so no independent-binomial confidence interval is assigned to that fraction. The boundary cannot be retuned after the audit. Failure of this gate remains a completed development observation with the gate marked false, not a runtime error or a reason to change the method. The observed development result is reported in Section 4.3.

Training BIC selects only among the six prespecified candidates; neither convergence nor minimum BIC establishes useful monitoring. Disjoint calibration and audit domains separate boundary selection from its assessment, but both streams remain development-

source data. Selecting a calibration percentile does not guarantee the audit fraction, and overlapping windows are not independent trials.

Every complete 256-request window of the retained external stream is a prespecified external-shift window. The detection-rate numerator is windows with score strictly greater than the frozen boundary; the denominator is all complete windows. Overlapping windows count separately. A terminal incomplete window is excluded from the rate, but its requests can still be routed by an alert from a prior complete window. The independent validation-audit false-alert fraction uses the same complete-window numerator and denominator rule.

External records remain in published file order after the applicable frozen quarantine. The first 256 requests use the fixed cascade. Windows then end at request 256 and every 64 requests thereafter. An alert routes exactly the next 256 requests through the transformer, without rerouting the records that generated it. Overlapping activations are unioned, terminal activations are truncated at stream end, and each request invokes the transformer at most once. This is offline replay with a prospective, future-only action rule. Its paired outcomes assess routing utility conditional on the observed source/domain shift and stream order, not a causal effect in a prospective field trial.

For that comparison, the reduction in false-negative rate is recall(policy) minus recall(fixed cascade) on the same NCSC gold positive rows. Bootstrap resampling uses the saved, already-routed outcomes; it never replays routing on resampled rows. The domain-clustered interval is conditional on the realized stream and order. It does not

capture cross-domain temporal dependence from shared routing activations or uncertainty over alternative stream orders.

The primary monitor claim remains limited to P(X). A GMM window alert is evidence of distributional departure under the chosen representation. It does not by itself establish harmful drift or a change in P(Y|X). Maximum mean discrepancy (MMD), population stability index (PSI), and controlled string perturbations are secondary checks and cannot satisfy H2.

3.10 External Evaluation Freeze

The external evaluation uses Mendeley Data repository Version 4 (PhishVN v3.1.0; Vu, 2026b). The repository metadata identifies Version 4 as a documentation-only update to Version 3: every data file is byte-identical, while the datasheet adds the completed inter-annotator label-audit results. The external population is the full published test split in published file order after only the common and source-specific quarantine rules, applied to the representation specified for each recorded attempt. Every retained record participates in label-blind routing. Outcome filters are applied afterward. The models are neither refit nor recalibrated.

The original pre-access procedure required the protocol and its inputs and procedures to be frozen and hashed. The data specifications covered the source contract, full-test population and ordering rules, routing and outcome denominator rules, and manifest specifications. The fitted-artifact specifications covered model artifacts, vocabulary, feature schema, fitted preprocessing, model parameters, thresholds, cascade band, GMM

and alert boundary. The freeze also covered statistical code and the software environment.

First access verified schema metadata, categorical values, license, provenance and encoding before applying the mapping and quarantine rules without model scoring. Realized preparation identities and manifests preceded prediction. The September 28 preparation reached the whole-study hold; both source scorers and all operational cells were unattempted in that original root. A missing or inconsistent required field remains a reason to stop rather than impute a replacement.

The publisher-URL amendment was adopted after the hold exposed preparation and label counts, including exclusions of gold and Tranco records lacking schemes in the raw field. It authorized exact url_norm inputs and preserved original raw cells, without reopening the original sources. That decision preceded model predictions under the amendment and was prediction-blind in that limited sense; it was not before test-data access. The subsequent full attempts repeated source evaluation. A later checkpoint proposal followed those predictions and partial operational execution and cannot inherit the earlier prediction-blind description. Operator authorization does not establish advisor approval or personal review of later implementation hashes.

3.11 RQ2 Analysis

The GMM is applied without labels to the full retained external stream. Prespecified windows determine the external-window detection fraction. The independent validation audit determines the reference false-alert fraction. Paired fixed-cascade and alert-policy outcomes are then computed for primary external strata. The error comparison uses a

registrable-domain-clustered 2,000-replicate percentile bootstrap. H2 is decided only by the four gates in Section 1.4.2.

Because the policy is evaluated on an observed ordered stream rather than a randomized intervention, its result is an evaluation of the specified routing policy. The analysis does not attribute any error change solely to the GMM alert or describe the replay as proof of causal remediation.

3.12 RQ3 Operational Replay

The primary benchmark manifest contains 10,000 URLs drawn without replacement from the PhiUSIIL group-test partition at a declared 1% phishing prevalence using seed 20260816. The declared prevalence is a measurement reference, not an estimate of production prevalence. Manifests at 0.1% and 5% are sensitivity analyses. The primary H3 replay uses the fixed cascade with no alert-triggered routing expansion. Future-only shift-period routing is evaluated separately under H2, and transformer-only worst-case load is reported separately.

The real-HTTP harness evaluates concurrency 1, 8, 16, 32, 64, and 128. Each setting uses 1,000 warm-up requests and 10,000 measured requests in each of five runs. H3 evaluates latency and errors at concurrency 64. The request-error denominator is all 50,000 measured concurrency-64 requests. Warm-ups are excluded. Success requires exactly HTTP 200 and a valid, identity-matched JSON response within 2,000 ms. Other status codes, invalid responses, timeouts and transport failures remain in the numerator and denominator. Pooled p95 uses the linear NumPy quantile over all individual measured concurrency-64 latencies, including failures, rather than averaging per-run p95 values.

The prospective manifest supplement sorts prepared candidates by stable source-row ID, samples each class without replacement, then permutes the combined selection. Separate PCG64 streams use seed components [20260816, 1, prevalence in basis points, purpose], where purpose distinguishes negative selection, positive selection and final order. Every repeat and concurrency uses that same order. The first 1,000 rows are replayed as warmup before the full measured manifest. An insufficient class count is reported without replacement sampling or a revised target. A canonical private-payload SHA-256 binds all selected record fields, including original raw URL spelling.

The HTTP supplement specifies one inference owner and a first-in, first-out (FIFO) queue for at most 128 waiting requests, excluding the active request. Each run starts a fresh service and client. Closed-loop workers time requests from submission through body reading and response validation, excluding unsent local backlog; there are no retries. This is a closed workload in the sense of Schroeder et al. (2006): each worker submits another request only after its preceding request terminates. The total deadline includes all connection, queue and processing time. Observed terminal latency is retained rather than clamped at the deadline. Admitted work continues after a client timeout. Between phases, a control request closes the previous phase's IDs to late admission and drains the owner queue before recording counters. Warmup failures are recorded without rerunning warmup. Physical invocation evidence comes from the first concurrency-1 run of the primary 1% manifest, designated before results; it is not selected from the most favorable repeat. Source/artifact authentication, hardware binding, separate worst-case workloads and execution controls were integrated for the study attempts recorded in Section 4.6.

The fixed schedule contains 125 cells in 25 five-repeat groups. Ninety cells evaluate the fixed cascade across three prevalences and six concurrencies; thirty evaluate transformer-only inference on the 1% manifest; five evaluate serialized live-shift replay. Each fixed or transformer-only repeat measures 10,000 requests. Each live-shift repeat measures the actual 8,701-row external stream after a separate 1,000-request warmup and phase reset.

The accepted operational matrix has a two-session structure. An all-or-none review admitted the complete first 72 cells from one retained attempt, after which cells 73–125 were measured in their prescribed order. The checkpoint rule was adopted after source predictions and partial measurement. Cell 73 was measured afresh in full rather than assembled from stopped and completed requests; Appendix C preserves the complete chronology.

All 125 accepted cells retain their prescribed ordinals, manifests and request/deadline rules. The primary H3 reference cell 1 and latency/error cells 21–25 belong to the first accepted session and are not pooled with replacement primary measurements. Descriptive group 71–75 spans the session boundary. Because session and fixed workload order are aligned, their effects are not statistically separable in this design.

For both the fixed cascade and the transformer-only model, external observed FPR on certified trusted-registry negatives must be <= 1%. In addition, the Tranco reference-negative alert rate must be <= 1% for each system as a mandatory secondary safeguard. Exact one-sided 95% Clopper-Pearson upper bounds are reported for all four rates. H3 retains the recall margin, invocation, latency and request-error requirements stated in

Section 1.4.3; Section 3.8 corrects the paired uncertainty used to adjudicate its recall margin.

3.13 Reproducibility and Integrity Controls

Reproducibility has three roles in this study: identifying the inputs and fitted state, reconstructing the reported arithmetic, and preserving the distinction between development and evaluation. Versioned contracts fix feature order, operating points, population rules and measurement denominators. Appendix C lists their exact identities and changes over time.

Development feasibility and test performance remain separate. The accepted structural and character artifacts entered evaluation unchanged. The GMM development run did not meet its reference false-alert requirement, so a favorable external-window response could not establish the original joint H2 claim. No primary numerical threshold is relaxed by a correction or later experiment.

The versioned research record is cited as Tallam (2026). Appendix A maps manuscript results to code and aggregate evidence; Appendix C records input and execution identities. Independent calculations verify counts, paired outcomes and latency summaries from saved observations. Such verification checks arithmetic and linkage; it is not a second investigator's empirical replication.

The secondary program examines representation shortcuts through a formatting-only comparator, training-label permutation sensitivity and an explicit assessment of source identifiability. It also includes seed sensitivity, Random Forest, MMD, PSI, controlled perturbations, exact McNemar tests and Holm-adjusted ablations. These analyses explain

model behavior and qualify interpretation while leaving the primary decision rules unchanged. Test-informed retuning is prohibited, and the report retains the full comparator set, including negative and mixed outcomes.

The September 21 secondary-analysis supplement records the remaining conventions before protected evaluation. Implemented score summaries include AP, ROC AUC, recall at observed FPR <= 1% without interpolation, Brier score and ten-bin expected calibration error (ECE). Binary operating-point decisions are supplied separately from probabilities because the cascade uses different stage thresholds. The four paired contrasts use the exact conditional form of McNemar's (1947) test with Holm's (1979) familywise adjustment. They compare Logistic-L1 with length-only and cascade with Logistic-L1 in the internal and external gold populations; Holm adjustment retains all four slots even if a contrast is unavailable. These nominal paired tests do not account for domain dependence and do not replace the primary clustered intervals. A source-only comparator requires collection-source identifiers independent of the outcome label. PhiUSIIL provides no per-row identifiers of that kind, so source identity is not inferred from phishing status; the resulting identifiability boundary is part of the secondary analysis.

The September 22 secondary-development contract specifies the comparator and tabular-model procedures separately from the primary models. MMD/PSI training references use the accepted GMM scaler and portable monitor probability and are constructed before validation rows are received. The procedure preserves the original domain-based calibration/audit allocation and strict 95th-percentile calibration rule, with a separate boundary for each comparator.

The tabular models comprise the five-indicator formatting model, five training-label permutation controls and the fixed Random Forest. Each label permutation uses a fresh PCG64 seed from 42 through 46; the Logistic-L1 solver seed stays 42, and validation labels are unchanged. Each model has a separate Clopper–Pearson (CP) validation cutoff. CPU singleton scoring must agree exactly with probabilities reconstructed from the serialized numeric state. No primary artifact or cutoff is replaced. Random Forest (RF) uses the direct-leaf arithmetic specified in the prospective correction described in Section 4.6. Section 4.5 reports the accepted secondary evidence.

Secondary analyses use the accepted model and operating-point artifacts without promoting a favorable comparator into the primary design. Saved outcomes retain their source and analysis role. Appendix C separates failed predecessors, accepted stages and later corrections.

The September 23 seed/probe supplement specifies a separate training entry for seeds 43-46, reusing the accepted training-only vocabulary. Seed 42 is not refitted. Training retains batches of 256, checkpoint AP scored in batches of 512, and the original stopping rule. Each epoch's ordered validation probabilities and AP are exposed for retention, along with every qualifying fitted checkpoint before later epochs or checks. All five weight sets use the same singleton secondary scoring procedure and the accepted stage-one model and historical cutoff. Each transformer receives its own secondary CP cutoff and cascade band; no primary artifact or operating point is overwritten. The historical seed-42 fit used its recorded earlier runtime, so these comparisons assess seed/runtime sensitivity rather than an isolated random-seed effect.

Probe replay uses the original validation audit stream and three separately transformed copies. Each copy retains the exact original URL, eligibility, changed status and row position, including ineligible rows and no-ops. The four primary detectors and GMM routing policy are scored, while GMM, MMD and PSI use their saved references and boundaries on complete 256-row windows at stride 64. Monitor and routing history start empty for each stream. Results are paired score, decision and alert changes, not correctness or adversarial-success rates: transformed strings do not inherit outcome labels. The saved-reference loader and development adapter check retained evidence, the original audit allocation and model/scaler identities without refitting or recalibrating.

The character-seed and label-free probe analyses retain every specified member. Their saved-evidence checks reconstruct checkpoint selection, calibration and decisions; the probe transformations and resulting measurements remain descriptive. The separate post-review controls in Section 3.16 add direct fit-boundary evidence for shuffled training labels.

Earlier character-seed stages were accepted together through a no-fit evidence review, with a separately executed probe correction. No seed was selected or dropped for its performance. Appendix C retains this sequence and the stopped predecessor; it does not imply that a later verification preceded the original development observations.

3.14 Ethical and Safety Considerations

The study treats URLs as inert strings and does not contact their destinations. The original versioned research release distinguishes licensed source records and model evidence from private execution files. For the post-review cohort, URL strings, domains,

row identifiers and record-level predictions remain private; the public supplement contains aggregate results, specifications, code and permitted file-level identities. This narrower release boundary preserves verification without exposing newly retained identities. The gateway returns alerts for review rather than performing automatic blocking or takedown.

Manual review is permitted only as separately reported post hoc descriptive error analysis. It cannot assign or override labels or change quarantine, inclusion, thresholds, features, models, gates, or hypothesis decisions.

3.15 Bounded Follow-up Design and Evaluation

3.15.1 Sequence and Scope

The follow-up proceeded from the initial evaluation to diagnostic investigation and bounded comparison. On October 1, 2026, after inspection of the initial results, requirements D and S were specified before retrieval of the additional benchmark CSV, fitting of the revised detector or measurement of the final client comparison. The extension was therefore development-informed, not specified before the original results. It added two distinct comparisons without replacing H1–H3, their thresholds or their completed decisions. Operator authorization did not constitute advisor or institutional approval; the original manuscript, experiment records and result package remain preserved.

Two mechanisms define the scope. Detection comparison D tests whether removing scheme spelling consistently from the structural representation improves external recall under a low-FPR constraint. Service comparison S tests whether worker-owned persistent

HTTP clients reduce tail latency relative to the original shared pool. The character transformer, GMM and future-only routing policy are not retrained or retuned. The extension does not test a new drift-remediation policy and cannot supply new support for H2. Each primary follow-up comparison uses a two-sided 97.5% interval, allocating the familywise error budget across two comparisons without assuming independence. The intervals are approximate bootstrap intervals, not exact guarantees of coverage.

3.15.2 Additional Benchmark and Eligibility

Hannousse and Yahiouche's Web page phishing detection dataset, Mendeley Data Version 3, supplies the additional benchmark (Hannousse & Yahiouche, 2021b). Its associated study addresses reproducible benchmark construction and feature evaluation (Hannousse & Yahiouche, 2020, 2021a). The retained publisher CSV contains 11,430 URLs, with 5,715 publisher-phishing and 5,715 publisher-legitimate labels. The file is distributed under CC BY 4.0. Only its URL index and explicit class strings are used. None of its 87 publisher-computed features, stored page objects, or live destinations enters this experiment. No URL is fetched. Publisher classifications remain reference labels, not independently adjudicated truth or the original PhishVN gold/certified tiers.

Metadata reports May 2020 collection, whereas Section 4.3 of the inspected preprint states March 2020. The analysis therefore refers to the 2020 benchmark without asserting a verified exact collection month. Its described source mechanisms include Alexa-seeded crawling and Yandex for legitimate examples and PhishTank/OpenPhish for phishing examples. These mechanisms differ in part from the development source but share upstream phishing feeds. The benchmark is neither a representative production prevalence sample nor a fully independent source mechanism. The local research-record

search found no previous use of this benchmark; that search is not proof of the absence of every historical exposure.

Eligibility reuses canonical-url-v1 and the original pinned Public Suffix List, including its private-suffix policy. Before any model prediction, the procedure quarantines invalid URLs, mixed-label canonical duplicates and registrable domains observed anywhere in the retained valid-source PhiUSIIL manifest or complete PhishVN publisher manifest. Both raw and publisher-normalized PhishVN fields across all splits contribute available domains. There are 50,832 unparseable field instances in that comparison source; they are not 50,832 missing records, because raw and normalized fields can differ in parseability. No guessed domain is assigned to a malformed field. Same-label canonical duplicates retain their first source-order occurrence. All raw rows and exclusion reasons remain preserved.

Admission requires at least 1,000 retained rows and 250 distinct registrable domains in each class. These are minimum-information requirements, not a power guarantee. Failure of either class would hold the whole final extension comparison rather than trigger a search for another favorable dataset. The admitted population contains all 8,622 eligible rows: 4,651 phishing records in 2,897 domains and 3,971 legitimate records in 3,377 domains. There are 6,273 distinct domains overall because one domain has distinct eligible records in both classes. Domain grouping retains this dependence. No balancing or outcome-based sampling is performed.

Figure 3.2 separates the roles of development, diagnosis and additional evaluation. Previously exposed evaluation outcomes inform the mechanism-directed design; the

unchanged training and validation partitions supply coefficient estimation and threshold selection. Retained full-source domain manifests are used only for overlap screening of the additional benchmark. They are not new fitting observations. The admitted benchmark supplies the final paired comparison and does not select the candidate or its operating point.

Figure 3.2. Source and partition roles in the bounded detection comparison. Arrows show permitted information flow; domain exclusion does not establish temporal or source-mechanism independence.

3.15.3 Transport-Neutral Structural Representation

The unchanged comparator is the accepted 25-feature Logistic-L1 model at threshold 0.2670846328466124. It is not refitted. The revised detector validates the original absolute HTTP(S) string, replaces only the scheme prefix with the fixed spelling http, preserves the remainder of the string and computes the original structural features except is_https. It therefore has 24 features. Scheme spelling is removed from all affected lengths, character counts, ratios and entropy, not only from the explicit indicator. This operation changes model inputs; it does not edit the retained raw URLs or describe the security of a live transport connection.

The candidate was fitted once using the unchanged 166,248-row PhiUSIIL training partition. StandardScaler used training data only. The LogisticRegression configuration retained L1 regularization, the saga solver, C=1, balanced class weights, an intercept, a maximum of 5,000 iterations, tolerance 0.0001 and seed 42. Numerical execution used one thread. The existing maximum-recall threshold rule was applied only to the original

validation partition of 32,695 records: candidate thresholds required a one-sided 95% Clopper–Pearson FPR upper bound at or below 1%, with the existing exact tie rules. The fit and operating-point artifacts were frozen before external scoring. The stopping policy permitted no solver, seed or tolerance search after nonconvergence or numerical-integrity failure.

The fit converged in 4,971 iterations and selected threshold 0.6815549262616749. On development validation, it detected 7,588 of 12,486 positives and alerted on 178 of 20,209 negatives: recall was 60.77%, observed FPR was 0.8808%, and its one-sided upper bound was 0.9968%. These quantities defined the operating point; they were not external efficacy evidence. The lower development recall is a measured cost of the revised fitted representation at that threshold. The comparison does not isolate the contribution of one coefficient or establish that all information removed by scheme neutralization was spurious.

Each eligible benchmark URL is evaluated in its original form and with one opposite-scheme companion, preserving the remainder. The candidate must produce exactly equal feature vectors, scores and decisions for the two forms. These companions are metamorphic representation checks, not independently labeled websites or evidence that changing a live site's transport leaves its phishing status unchanged. Both models use singleton sklearn prediction with the existing independent float64 reconstruction audit. Models and thresholds are not selected on the additional benchmark.

3.15.4 Detection Estimands and Uncertainty

The primary D estimand is candidate recall minus unchanged-comparator recall on the original eligible URL strings. D requires a point gain of at least five percentage points, a positive lower endpoint of the two-sided 97.5% domain-cluster interval, and candidate observed FPR at most 1%. Exact candidate scheme invariance is a separate required representation-conformance check, not an additional efficacy endpoint. The effect target is a follow-up engineering requirement rather than an original hypothesis or a literature-derived threshold. All conditions must hold; ranking or calibration improvements cannot substitute for them.

The analysis resamples the 6,273 lexicographically ordered registrable domains with replacement in 10,000 paired PCG64 replicates with seed 20261001. A selected domain contributes all of its rows with the domain's sampled multiplicity. Recall and FPR are row-weighted ratios within each replicate, not equally weighted averages of per-domain rates. Undefined positive or negative denominators are counted and excluded only from the corresponding interval calculation. The same resamples produce the candidate FPR interval. The observed 1% decision rule is distinct from uncertainty about population risk. Confusion matrices, precision, ROC AUC, average precision, Brier score and ten fixed equal-width calibration bins are all retained. Empty bins and zero-alert precision are undefined where appropriate, not assigned fabricated values.

3.15.5 Service Comparison and Its Boundary

The service comparison varied client connection ownership while retaining the HTTP endpoint, input sequence, response contract, two-second deadline, no-retry rule and latency boundaries. A shared persistent pool was compared with one persistent client per

closed-loop worker. Both topologies were evaluated on a no-model response control and the unchanged singleton structural detector; neither workload invoked the transformer. The control assesses the client/service path without classifier work, but does not independently identify each scheduling, queueing or connection-pool cost. The fixed synthetic manifest contains 10,000 unique occurrence IDs with reserved .example hosts, alternating schemes and distinct paths. It has no outcome labels or asserted phishing prevalence.

The schedule specifies concurrency 1 and 64, ten paired repetitions per workload/concurrency combination and a fresh separate-process service for each arm: 80 arms in total. Odd pairs use shared then worker clients; even pairs reverse that order. Each arm has 1,000 warmups, 10,000 measured requests, phase drains and retained request-level outcomes. The endpoint uses one serialized scoring owner. An environment, startup or cleanup interruption stops the remaining schedule; partial results are preserved rather than selectively pooled or retried for a favorable outcome.

The primary S estimand is the median of ten paired per-run p95 ratios, worker divided by shared, for the structural workload at concurrency 64. Ten thousand paired run-bootstrap resamples use PCG64 seed 20261002 and a two-sided 97.5% percentile interval. S requires median ratio at most 0.8, upper interval endpoint below 1, pooled successful worker-request p95 at most 200 ms, fewer than 0.1% errors among all 100,000 primary worker requests and exact paired response agreement excluding request IDs only. Admission-sequence differences remain part of this strict agreement condition. Prediction-field agreement is reported separately so that response-order differences are

not misrepresented as changed model decisions. With only ten pairs, bootstrap uncertainty has limited resolution.

Service timing used a reserved host with sampled power, thermal and workload observations. Successful, failed and all-request latency denominators, throughput, physical counters and process exits are retained separately. These controls describe the measurement conditions, not a continuously idle or disturbance-free machine. The synthetic workload and altered client cannot replace the original H3 measurements or establish a transformer speedup; operational details are in Appendix C.

The complete S schedule followed an interrupted predecessor. Its replacement rule was specified after partial nonprimary observations, before any primary structural concurrency-64 arm had begun. All 80 final arms use the unchanged model, thresholds, deadlines and response contract; no predecessor arm contributes to the final estimate. Appendix C documents the amendment and retained interruption.

3.16 Post-review Diagnostic and Routing Comparisons

The October 12 post-review study addresses the mechanisms raised by the original evaluation without reopening model selection. Its specification fixes four linked tasks before any new fit or cohort prediction: F, a frozen-model formatting intervention; R, a domain-balanced retrospective routing comparison; C, actual routed-computation measurement; and P, five audited shuffled-label fits. Every task requires eligible inputs before the study begins. Original H1–H3, D and S remain separate, and no new support for H2 is inferred from these tasks.

3.16.1 Frozen-model Intervention and Domain-balanced Sampling

F uses all 8,701 previously evaluated external strings. It preserves labels, the accepted structural weights and scaler, and threshold 0.2670846328466124. Each retained spelling is compared with forced http: and https: prefixes, preserving every character after the first colon. Each form is scored once; the original form must reproduce its retained result. Paired features, scores and decisions identify the spelling effect on the frozen model. Changing a string is not an intervention on a live website and does not isolate a single feature, because the prefix enters several structural features.

R uses retained PhishVN publisher train/validation records (Vu, 2026a, 2026b), not the publisher test partition. Positive eligibility requires the tinnhiemmang source, gold tier and phishing label. Negative eligibility requires the tinnhiem_org or tinnhiem_web source, gold tier and benign label; label_source must be present. The unchanged canonicalizer rejects invalid URLs, contradictory canonical labels and hostname disagreement. Domain exclusions cover publisher test, cross-split domains, PhiUSIIL, D and the complete discovered historical exposure roster. The cohort is locally domain-disjoint, not established as campaign-independent, temporally new or globally unexposed.

The fixed eligible yield is 1,392 positive and 1,262 negative domains. Every eligible domain contributes one representative: the URL with the smallest fixed-prefix SHA-256 ranking, with deterministic ties. A separate domain hash fixes evaluation order. This domain-balanced retrospective estimand differs from the original row-weighted external study. Exact publisher url_norm is the model input; canonical identities serve only

eligibility and overlap checks. No predictor output determines inclusion, no new domain is added after scoring, and these labels do not select a model, threshold or gate.

3.16.2 Routing Comparisons and Paired Inference

R scores the accepted structural model and character transformer once per record using their frozen singleton scoring convention and thresholds. The transformer threshold is 0.033976949751377106. Saved component decisions define structural-only, always-transformer, the original zero-width cascade, a symmetric half-width 0.20 gate, and a directional 0.20 band below the structural threshold. The symmetric rule is the selected primary revised gate; the directional rule is a secondary comparison. Both were selected from the bounded development replay in Section 4.8.1 before R predictions. Fixed identity-hash controls select the same number of requests as each revised rule without using correctness. These controls are finite-batch comparisons, not online routers.

The three primary R statements concern original-cascade recall noninferiority to the transformer, symmetric-cascade recall noninferiority to the transformer, and symmetric-cascade recall improvement over structural-only accompanied by observed FPR at most 1% and calls at most 30%. Noninferiority retains the two-percentage-point margin. Each contrast uses the conservative two-sided 95% discordance envelope in Section 3.8; Newcombe's (1998) method 10 is a prespecified sensitivity. These are individual, conditional intervals rather than simultaneous coverage of all comparisons. Counts, logical calls, negative and positive repairs/harms, and ranking measures are retained for every system.

The sample size is the fixed eligible yield, not a target enlarged until a decision passes. Planning at 1,392 independent positive pairs gives approximately 98.25% power under the exposed internal symmetric-rule discordance proportions. Sensitivity falls to 53.76% at equal recall and 5% total discordance, illustrating that the 90% planning target is scenario-dependent. The actual decision depends on observed paired outcomes and assumptions, not planning power. Positive source-event-date groups are also resampled in 2,000 fixed PCG64 replicates as a descriptive dependence sensitivity. Date groups are not verified campaigns, and a degenerate bootstrap does not supersede the exact envelope.

3.16.3 Actual Detector Work and Audited Controls

C fixes 10 positive and 990 negative R strings in cohort identity order before seeing their scores. The 1% mixture is a workload reference, not estimated traffic prevalence. Six scoring paths—structural, transformer, original cascade, symmetric, directional and symmetric matched-count control—each execute five repetitions of the same 1,000 measured requests at concurrency one. Each arm first executes 100 excluded warmups. Five cyclic rotations of the six-path order are not fully position-balanced. Models load outside the timed region; request timing includes input processing, gate work and any actual expert scoring, but excludes evidence writes and admission checks. Counts distinguish measured requests from warmups. This is not an HTTP service benchmark.

C retains p50, p95 and p99 for each arm, the summed request-scoring times, scores, decisions and actual stage starts/completions. Within each repetition, summed scoring time is divided by the structural-path total; the five paired ratios are summarized by their median and range. No run-level confidence interval or production-capacity estimate is claimed. Output parity is checked against R's saved component scores. An always-

transformer path exercises expert computation separately from natural routing; it is a cost reference, not representative selective traffic.

P fits the prescribed Logistic-L1 pipeline once for each seed 42–46 on the unchanged training identities, shuffling only the ordered training label vector. At the estimator.fit boundary, the retained receipt binds that exact consumed vector, row order, class counts, permutation and transformed input. Reloaded fitted state and saved validation predictions are checked independently. The unchanged 32,695-row validation set has 12,486 positives and 20,209 negatives; its true labels select each descriptive operating point with the existing Clopper–Pearson constraint. Thus, reported validation recall is not independent test efficacy. The five runs are diagnostic controls, not a formal permutation significance test (Ojala & Garriga, 2010). They neither select a favorable seed nor replace the historical fits.

Chapter 5—Discussion and Conclusions

5.1 Direct Answers to the Research Questions

The primary investigation answers RQ1–RQ3 through completed comparisons and explicit numerical criteria. After correction of paired uncertainty, eight of 22 original component checks pass: five for H1, one for H2 and two for H3. Each joint hypothesis requires all its components, so none is supported by a majority count. The later studies explain the decisions rather than replace them: D tests a revised representation, S tests client connection ownership, F isolates scheme spelling in the frozen model, R evaluates bounded routing changes on a domain-disjoint cohort, C measures their actual

computation, and P audits shuffled-label fitting. Together, they follow the same argument from design through diagnosis to measured modification.

RQ1: What incremental value do structural URL features and character-level representations provide under registrable-domain-disjoint and external evaluation?

Structural features add substantial recall over length alone: 64.07 percentage points internally and 76.81 points on external gold positives. The internal domain-clustered interval is [59.99, 67.63] points; the corrected external paired envelope is [57.27, 87.19] points, conditional on independent labeled pairs. Internal structural FPR is 0.7747%, satisfying the 1% operating constraint. External transfer is different: certified FPR is 90.5487% for both Logistic-L1 and the fixed cascade, compared with 9.4513% for length-only. The validation-selected cascade band routes zero evaluated records to the character model, making its decisions identical to Logistic-L1 and its incremental recall gain exactly zero. Transformer-only adds a descriptive internal recall gain but has 99.2791% certified external FPR. These measurements answer RQ1 in two parts: structural information improves recall in the paired comparisons, while the selected operating points do not retain external specificity and the fixed routing band contributes no additional detections. H1 is unsupported because external FPR and incremental cascade benefit are required alongside the demonstrated structural gain. Source and representation differences motivate the subsequent diagnostics; these comparisons alone do not isolate a single causal source of degradation.

The representation studies refine this answer. D achieved exact feature, score and decision invariance across all 8,622 companion pairs, but its separately fitted detector lost

184 positive detections while removing 251 false positives. F then held weights, scaler and threshold fixed: forcing HTTPS removed 2,211 false positives while losing 47 of 69 gold detections. These experiments answer different questions. D establishes that a consistent scheme-neutral contract can be implemented and tested; F demonstrates that changing spelling alone can move the accepted classifier across its decision boundary. Neither supports treating a lower false-positive count in isolation as a successful detector repair. In R, widening the gate added 18 new false positives and no positive detections, so the external comparison supplies no benefit that would justify that routing change.

RQ2: Can GMM-based monitoring detect an external source/domain shift and guide escalation without exceeding the low-FPR operating constraint?

The GMM detects external departure in 115/132 windows (87.12%), satisfying the >=80% sensitivity requirement. Its 28/252 reference alerts (11.11%) exceed the <=5% false-alert requirement. Acting on those alerts routes 94.8512% of the external stream to the transformer and increases certified false positives by 218, producing policy FPR of 2479/2497 (99.2791%). Gold recall remains 69/69 for both the fixed cascade and routed policy. The baseline therefore has no observed gold false negatives for escalation to correct in this sample; the observed paired false-negative-rate reduction is zero. No positive population lower bound is established for this stateful replay. RQ2 is answered by the distinction between detecting a change in the input distribution and improving the decisions made after that change. The monitor demonstrates distribution sensitivity, but this escalation policy adds false positives and computation without a measured gold-recall benefit. H2 is unsupported because the reference-alert, policy-FPR and false-negative-reduction conditions are unmet, even though external-window detection passes.

RQ3: What detection, escalation, throughput, and latency tradeoffs determine whether the fixed cascade is viable inline?

The fixed cascade achieves two operational efficiencies: zero transformer calls among the 10,000 reference requests and zero request errors among the designated 50,000 concurrency-64 requests. These are the two passing H3 components. Both detectors find all 69 gold positives, but a corrected lower confidence bound of -6.1533 percentage points does not establish the required -2-point recall margin. The data show agreement on these cases, not a sufficiently precise guarantee about future cases. Pooled all-request p95 is 364.30 ms against the <=200-ms requirement, and both systems exceed the certified-negative FPR and Tranco alert limits; each flags all 1,163 Tranco controls. RQ3 also requires the throughput tradeoff: in the 1% fixed workload, attempted throughput is approximately 1,244–1,272 requests/s at concurrency 8 but 399–503 at concurrency 64. Increasing concurrency therefore does not provide proportional throughput growth in this measured service path. H3 is unsupported because external alert burden and tail latency exceed their budgets and the recall margin is not established, despite demonstrated call economy and primary request reliability. The next comparison tests whether changing client connection ownership can improve the service path while keeping the scorer fixed.

The expanded R cohort answers the precision question separately. Its original and symmetric cascades both satisfy the prespecified recall-only noninferiority comparison to the transformer, with a lower 95% bound of -0.3143 percentage points on 1,392 positive domains. That finding is conditional on independent labeled pairs and the source frame. It does not supply specificity, explain a production workload, or retroactively turn the original H3 result into a pass.

The service follow-up identifies an addressable part of the operational cost. Worker-owned clients reduced median paired p95 by 79.68% on the synthetic structural workload at concurrency 64; pooled successful-request p95 was 72.00 ms, with zero errors in 100,000 primary worker requests. Comparable predictions agreed, but admission-order differences and one shared-client timeout prevented strict response equivalence. C answers the separate detector-cost question using actual retained URL strings: the symmetric gate invoked the transformer on 17 of 1,000 measured requests and required a median 1.646 times the structural scorer's summed processing time. Its median run p95 was only 0.223 ms, while p99 was 6.769 ms. Rare expensive calls can therefore leave p95 almost unchanged while altering the upper tail and total work. Neither C nor the structural-only S comparison establishes optimized transformer-service capacity.

5.2 End-to-End Argument and Contribution

The central engineering question is whether an additional computation repairs a decision that the cheaper model gets wrong. The original external outcomes contain 2,261 shared false positives among 2,497 certified negatives. No selector of those unchanged binary expert decisions can reduce observed FPR below 90.5487%, even if the selector knows the labels. This is an offline feasibility bound for those decisions, not an impossibility theorem for new features, thresholds or models. It explains why routing more of the same examples cannot solve that particular specificity problem. The bounded revised-rule experiment tests this reasoning beyond the original rows: on R, both selected gates introduce 18 false positives without repairing a structural error.

The representation comparisons identify a different intervention point. F establishes that scheme spelling can substantially alter frozen structural decisions, including both correct

detections and false alarms. D removes the scheme pathway from every affected feature, rather than deleting only is_https, and verifies exact invariance. Its detection tradeoff shows that stable representation and low-FPR discrimination are separate requirements. Consequently, the appropriate design decision is to retain the tested conformance property as an input contract while requiring new labeled evidence before claiming that it improves deployment performance.

The computation and service comparisons locate two distinct costs. C measures real expert execution and shows that widening a cheap gate increases processing time without a detection return on the expanded cohort. S holds the structural scorer fixed and demonstrates a large client/service-path latency reduction through persistent client ownership. The results do not collapse into one generic speedup: one concerns work performed by the detector, the other client-observed HTTP latency under a specified load and response contract. A systems designer can use this distinction to decide whether to change routing, representation or transport organization, rather than attributing every delay to the model.

This work extends an existing research discussion, rather than claiming priority for normalization, cross-source evaluation or calibrated cascades. Ahamed et al. (2026) already examine robustness and a separate PhishTank/Majestic evaluation; Rashid et al. (2024) and Tsai et al. (2024) address transfer and dataset bias. Li et al. (2021) and Alajaji (2026) establish cascade and selective-deferral precedents. ExpertFusion's author repository describes fixed URL encoders, lexical features, prior-specific experts and calibrated routers using unlabeled target statistics (aI-area, 2026; Hussain et al., 2027). The distinction established here is the measured connection among same-example expert

repair, a fixed risk constraint, actual invocation cost and a separately controlled HTTP intervention. The original algorithms are established; the empirical contribution is the tested decision logic and its boundary conditions in this URL-only gateway.

The research scope remains the URL-only gateway in Sections 1.4–1.5. The original 22 checks and 25 operational groups answer the stated representation, monitoring and inline-execution questions. Later comparisons explain their mechanisms and test specific modifications under separate designs. In particular, the larger R cohort establishes recall-only noninferiority under its stated sampling assumptions; it neither revises the original 69-case result nor supplies new support for H2. No original threshold, model or joint decision is replaced after observing a follow-up outcome.

5.2.1 Technical Value of the Completed Investigation

The technical value lies in several linked results that can guide another implementation. The study quantifies an error-overlap floor before changing a gate, tests the selected gate on different domains, and measures the cost of the work it actually invokes. It distinguishes a controlled input-spelling effect from a representation/refit/calibration intervention. It also shows why exact agreement on a small positive sample needs a nondegenerate paired interval, and why equivalent predictions need not imply an equivalent stateful HTTP response. Table 5.1 connects each result to a design decision rather than treating artifact volume as a contribution.

Table 5.1. Technical contributions, supporting observations and reusable value.

For representation research, F provides direct evidence of a frozen classifier's sensitivity to the scheme prefix. Moving from the retained spelling to HTTPS reduces certified FPR

from 90.55% to 2.00%, but recall falls from 100% to 31.88%. The threshold, labels and all characters after the first colon remain fixed. Several features change together, so this is not a causal estimate for the is_https coefficient alone. D complements that result: its consistent input contract achieves exact invariance after refitting, yet still misses the required external risk and recall-gain conditions. Together, the studies establish why a representation repair must be tested against both classes and why conformance alone cannot establish detection efficacy.

For cascade research, the original gate is an understandable optimum of its validation objective: minimize calls among feasible bands, then choose the narrower band. A zero-width band selects three validation ties and no original evaluation rows. The development-selected symmetric alternative repairs enough internal decisions to add 34 detections and remove 40 false positives, but that benefit does not carry to R, where it adds 18 false positives and no detections. C then measures the additional computation rather than inferring it from call counts. These results identify a concrete failure of benefit transfer from development to the expanded cohort and show why making a second stage active is not, by itself, an improvement.

For adaptive systems research, the study separates three propositions often compressed into one: the input distribution changed, the monitor reliably distinguishes that change from its reference, and escalation improves the decisions. The external GMM window gate passes, the original reference false-alert gate fails, and the downstream policy adds false positives without gold recall gain. Expanding transformer use to 94.8512% of the external stream is therefore not evidence of successful adaptation. The preserved future-only routing and the five matching live/offline traces establish what the policy executed;

they do not turn its adverse decision effect into a software malfunction or a favorable scientific result. The useful contribution is a concrete evaluation of the action that follows an alarm.

For service engineering, the original matrix establishes the inadequacy of inference-call count as a latency proxy. With no transformer calls in the primary reference cascade, pooled all-request p95 was still 364.3004101 ms at concurrency 64, above the 200-ms target. The matrix also records transformer-only errors at higher concurrency and distinguishes terminal latency, client-phase throughput, physical attempts and drain time. Those original measurements did not isolate the source of the overhead. The later paired client comparison narrowed the explanation by holding the scorer constant and observing a similar high-concurrency improvement in the no-model control. It supports the value of the client-topology intervention without uniquely separating connection-pool, queueing, validation and runtime-scheduling costs.

5.2.2 Engineering Work and Reusable Research Artifacts

The reusable artifact links deterministic source preparation, registrable-domain allocation, training-only fitting, validation-only operating-point selection, two-stage inference, windowed monitoring, future-only routing and real-HTTP measurement. Explicit source and execution identities connect these components to the reported analyses. A reader can trace a research question to its table, the table to an aggregate file, and the aggregate to the corresponding source and run lineage. Appendix A supplies the reproduction map. This traceability is useful because it exposes both the scientific estimand and the implementation conditions that produced each result.

The complete original matrix retains 1,243,505 measured request attempts, including all 1,901 request errors. Each of its 25 groups has a specified role and denominator, so a descriptive total never replaces the designated primary H3 cells. The post-review studies likewise retain all five P seeds, all three F spellings, every R system and all 30 C arms. File-level linkage and separate arithmetic checks make the findings inspectable; empirical value comes from what those observations establish, not from the number of checks or files.

The evaluation structure can be reused with a newly declared operating population or different detectors while retaining the separation among model selection, routing, labeled evaluation and service measurement. Its results identify concrete tests for subsequent work: whether external calibration can satisfy the intended risk ceiling, whether selected requests obtain a conditional benefit from escalation, and whether service optimizations preserve the required response semantics. These questions follow from the observed system, not from a claim that all URL cascades must exhibit the same behavior.

5.2.3 What the Engineering Iteration Adds

Table 5.2 separates the original joint hypotheses, their passing component checks and the later comparison requirements. A successful representation property contributes useful evidence without establishing a supported detection hypothesis. Similarly, an external-window alert demonstrates monitor sensitivity; the usefulness of the routing action requires its own evidence.

Table 5.2. Decisions at the hypothesis, component and follow-up levels.

The representation comparison converts a diagnostic possibility into a directly testable design property. The initial evaluation exposed a mismatch between within-source performance and external operating behavior. Scheme spelling was one plausible contributor because it entered both an explicit indicator and several string-composition features. The candidate removed that pathway consistently, used the original development partitions for fitting and threshold selection, and was evaluated on a separately admitted benchmark. Exact invariance confirmed that the intended transformation was implemented. The paired detection results then showed that this property alone did not provide the required recall gain or low-FPR operating point.

The technical lesson is specific. Dropping is_https alone would leave scheme information in lengths, counts, ratios and entropy. Applying a common prefix before computing every affected feature, while excluding the indicator, removed the prescribed scheme dependency. Exact paired features and predictions verified that property without contacting any destination. At the two frozen operating points, the candidate yielded 251 fewer false positives and 184 fewer detected positives. These paired counts make the cost of the modification explicit: improved conformance and fewer false alerts were accompanied by lower recall, and D remained unmet.

Behavioral preservation is an explicit object of evaluation across these comparisons. D defines preservation as equal features, scores and decisions under a scheme companion transformation. S distinguishes request-correlated prediction equality from exact response equality including admission sequence. R and C connect saved detector outputs with actual computation under the same singleton scoring convention. Specifying these

relations before their respective comparisons makes an achieved property interpretable without extending it to an untested contract.

The service iteration supplies a complementary positive result. When the initial latency budget was not met, the investigation separated no-model HTTP work from structural scoring, changed only client connection ownership and measured the full paired schedule. The measured improvement is large and consistent across all ten primary pairs, not a best-run selection. At the same time, observing prediction agreement alongside admission-order disagreement exposes two different meanings of behavioral preservation. The resulting engineering guidance is to measure the client path and specify whether ordering is a required externally visible property; neither model-call economy nor prediction equality alone proves the entire service contract.

5.3 Implications for Practice

Validation feasibility is not external safety. Before using a URL detector for inline alerting, an operator needs independent negative-source evidence at the actual intended operating point. Gold-positive recall alone can be perfect while nearly every trusted external URL is alerted. Tranco controls provide an additional alarm about generality, but popularity must not be relabeled as verified benign truth (Le Pochat et al., 2019).

Selective inference requires a useful expert on the cases selected by the inexpensive gate. The wider gate supplied internal repairs but no positive or negative repair in R; its added computation was measured in C. A routing evaluation should therefore report membership, error repairs, newly introduced errors and actual work together. For distribution-triggered escalation, an input-distribution alert is only the first step: the

original policy still needs to improve errors under the same specificity constraint. The present monitor passes external-window sensitivity but its action fails that requirement, which explains the unsupported H2 decision.

Service budgets require end-to-end measurement because queueing, validation, transport and serialization remain even when the expensive model is not invoked. The paired follow-up demonstrates that client connection ownership can materially affect high-concurrency latency without changing the structural scorer. It also shows why response semantics must be specified separately: comparable prediction fields agreed while admission sequence generally differed. Closed-loop throughput, terminal latency, service-side attempts and timeout/drain behavior should remain distinct measures. These observations guide design for the tested service, but do not establish production arrival-rate capacity, cross-hardware performance or suitability for automatic blocking.

The follow-up representation contract can be reused wherever publishers supply different HTTP(S) spellings, but invariance is not a substitute for labeled specificity evidence. On this benchmark, fewer false alerts coexist with fewer detected positives, and the 1% target remains unmet. Deployment decisions must therefore require both a stable input contract and a validated operating point in the intended population.

5.4 Limitations and Threats to Interpretation

First, all source labels are publisher reference classifications, not new contemporaneous forensic judgments. The original external recall analysis has 69 gold-positive domains, while its 2,497 certified negatives occupy only 234 domains. The revised 95% envelope for the original identical positive predictions is [-6.1533, 6.1533] percentage points. R

separately evaluates one representative URL in each of 1,392 positive and 1,262 negative domains, yielding a much narrower interval. One URL per domain changes the estimand and removes repeated-domain weighting, but does not prove independence across campaigns. Its recall-only conclusions remain conditional on independent labeled pairs; neither cohort represents global traffic prevalence.

Second, domain separation and source-mechanism separation are different questions. Per-row development metadata do not identify collection provenance independently of class, so a source-only classifier cannot be constructed without circular assumptions. F identifies a spelling effect on fixed model decisions, not the complete cause of source transfer failure. The five post-review P fits directly establish which shuffled labels entered each estimator, but their validation ranking still varies. True validation labels also select the reported operating points. These diagnostics do not authenticate historical fits, establish leakage absence or support a permutation-test p-value from five seeds.

Third, the interpretation depends on the information available when each methodological decision was made. Publisher-url_norm adoption followed preparation and label-count inspection; the checkpoint rule followed source predictions and partial measurement. These changes preserve the models, thresholds and decision rules but depart from the original input-timing and single-session intentions. Earlier analyst exposure and repeated source scoring also distinguish the realized evaluation from an unseen, one-lifetime-pass test. The preparation hold, excluded attempts and stopped requests remain preserved so that the accepted results can be assessed against that chronology.

Fourth, P(X) monitoring cannot establish harmful concept drift or causal improvement. Windows overlap; shared policy activations introduce dependence beyond individual domains. Bootstrap contrasts are conditional on the realized stream, and no routing is rerun within a bootstrap. The single published order is not a distribution of production stream orders. Label-free probes neither inherit correctness labels nor prove semantic preservation, adversarial resistance or real-world robustness.

Fifth, the operational estimates describe one host, a fixed workload order and two accepted sessions. Thermal, cache and elapsed-time effects may covary with workload and concurrency, and sampled condition checks do not separate those effects. Closed-loop client timing measures requests actually issued; it excludes unsent backlog and therefore answers a different capacity question from an open-loop production arrival process. Process-exit and saved-evidence checks establish execution and arithmetic integrity. Independent empirical replication would require a separate measurement study.

Sixth, the follow-up comparisons have limits specific to their data and execution. The additional benchmark is retrospective and publisher-labeled, shares upstream feed mechanisms, and has inconsistent collection-month descriptions in the metadata and inspected preprint. Excluding observed domains does not establish temporal or source-mechanism independence. The candidate was fitted and calibrated on the same development partitions used earlier; it is a separately evaluated modification, not untouched model development. Metamorphic scheme companions test representation behavior rather than the semantic equivalence of live websites. D's domain-cluster intervals remain conditional on the admitted benchmark.

S estimates a client-topology effect on one host with synthetic strings, a singleton structural scorer and ten primary run pairs. It does not estimate transformer-service improvement, field prevalence or open-loop capacity. C supplies a separate real-string detector benchmark at concurrency one and a designed 1% mixture. Its five cyclic rotations of six paths are not fully position-balanced, and its summaries carry no run-level confidence interval. The two studies therefore cannot be pooled into a single end-to-end speedup. Sampled host observations document recorded conditions rather than exclude every disturbance.

5.5 Future Research, Separate from This Study

Subsequent detection research should use independently adjudicated, time-indexed positive and negative sources to separate label, source and temporal effects. The observed shared-error floor motivates changed discriminative information or operating-point development, rather than more routing among the same binary decisions. Any such change requires development evidence separate from its final evaluation. The completed F, D and R comparisons supply starting measurements and rule out the tested scheme-only and gate-only remedies for the declared joint requirement; they do not justify a claim about every possible normalization or expert architecture.

Operational follow-up should cross session and workload order, evaluate open-loop arrivals and isolate queueing, scheduling and serialization costs. The response contract also warrants explicit design: an application requiring stable admission order poses a different requirement from one requiring request-correlated prediction equality. The completed comparisons provide measured baselines for those tests; they do not justify revising the present decision rules after observing the outcomes.

5.6 Conclusion

The study answers RQ1–RQ3 through completed original measurements and mechanism-directed comparisons. Eight of 22 original component checks pass, but H1–H3 remain unsupported under their joint requirements. H1 combines a real structural recall gain with external specificity loss and no original cascade gain. H2 detects external distribution departure, but its routing action adds false positives without recovering a gold miss. H3 demonstrates call economy and primary request reliability while missing the original latency, external-risk and recall-margin requirements. The post-review R cohort separately establishes recall-only noninferiority to the transformer on 1,392 positive domains, conditional on the stated assumptions; it does not change those original decisions.

The modifications and diagnostics produce concrete engineering knowledge. D achieves exact invariance on 8,622 paired inputs while quantifying the detection cost of its refitted representation. F demonstrates why a scheme-prefix change that removes thousands of false alarms is still unacceptable when it loses 47 of 69 positive detections. R shows that the selected wider gate offers no measured external repair; C quantifies the additional work and exposes an expert-driven p99 tail hidden by the cheaper p95. P adds direct fit-boundary evidence for all five new shuffled-label controls. S demonstrates a 79.68% median paired p95 reduction for the tested structural HTTP workload, with prediction agreement on all 99,999 comparable pairs, while preserving its unmet strict-response requirement.

CyberSentinel makes these findings traceable from input representation through score, routing decision, physical model execution and HTTP response. The resulting

contribution is a reproducible empirical account of when a component property becomes a useful system improvement and when another constraint prevents that inference. It gives a researcher specific measurements for deciding what to change next: conditional error repair before wider escalation, labeled operating-point performance alongside invariance, and actual detector and client-path costs rather than call counts alone. The narrative is therefore one of design, evaluation, diagnosis and measured comparison, with every favorable and adverse result retained in its proper scope.

Appendix C—Methodological Chronology and Execution Record

This appendix preserves source identities, contract history, stopped predecessors and the sequence of accepted corrections. The main methods retain changes that affect interpretation: information exposure, input representation, model selection, the two-session operational structure and the timing of later comparisons. Administrative records establish linkage and chronology, not advisor approval, empirical replication or predictive merit.

PhiUSIIL, UCI Machine Learning Repository dataset 967, is the sole source used for fitting, validation, and the internal held-out evaluation (Prasad & Chandra, 2024b). The source is distributed under CC BY 4.0 and is associated with DOI 10.1016/j.cose.2023.103545. The preserved archive's 256-bit Secure Hash Algorithm (SHA-256) digest is 0a639fd03aea6308c5b1c10c92aa23c2ce1505447a9137271865cd0badc9a59a. The embedded comma-separated values (CSV) file's SHA-256 is a236549cd369cd80bd478ff8e1779cbf44c58d5c3f79f7a51a1adbed7d06d1c6.

Registrable domains are computed with a commit-pinned Public Suffix List whose SHA-256 is 65365c4c9a4a6f746d53aadc758ab6b08aa10bb1379fea8ac353e381bca4b62e. Retained domain groups are ranked by SHA256 of the fixed allocation identifier, the seed 20260816, and the domain. Hamilton largest-remainder apportionment converts target shares of 70%, 15%, and 15% to integer counts for train, validation, and group test. Ties are ordered train, validation, group test. The allocation is label-blind and is not stratified, rerolled, or rebalanced.

The historical protocol v1.4 froze the first RQ1 machine-readable feature contract as rq1-baselines-v1, SHA-256 594a66769dee3bf23c4133020dcf9b7d57c105590e5007832ac4249def6a33d4. Its matrix SHA-256 was 2c2956e7cf958f9d2d948a2b1b665e214e12b175d84e4b73c766cc0a6e3be4de. The active baseline contract, frozen under protocol v1.7, retains the same feature vector in rq1-baselines-v2, SHA-256 05d6d0831def7d26448c8dbdc8117800ea2448cdfc2aca2ad95489f22d2d11ba; its matrix SHA-256 was ebd5b9f90157d6d21e8f22b7e1937c16dbaf60a577ec4f55cefc4709b604caef. The extractor returns an ordered 25-element float64 vector. Each value has one of three stated bases: the supplied URL string, urlsplit component text, or the existing IDNA-normalized ASCII hostname. It does not fetch a destination or substitute a canonical URL for the supplied string. The contract fixes feature order and input interpretation for the evaluated extractor.

The original v1 pipeline used liblinear, tol=1e-8, and a penalized synthetic intercept with intercept_scaling=1.0. Its full-feature fit reached max_iter=5000 and stopped as nonconverged. A later tolerance observation was not accepted because its command and executed configuration could not be reconstructed. The v1.5 SAGA diagnostic then stopped on a platform scoring warning. The v1.6 training-only diagnostic added a prespecified scoring audit and passed its reproducibility checks; it established training feasibility, not generalization. Protocol v1.7 froze that SAGA configuration before the validation run.

Protocol v1.8's rq1-transformer-cascade-v1, SHA-256 aeaa84534c4cadf0459cf6d2f010dc802684d4801cce563ce18242f36359fb54, remains byte-preserved with status superseded_unrun. Protocol v1.9 froze rq1-transformer-cascade-v2, SHA-256 686c0d86b33b8a6c2e09cd6e174003db0bd2f7c30b087faf5470e6a270524213. The v2 contract retains its freeze-time status frozen_not_run; dated execution records describe subsequent work. Version 2 changed only contract identity, schema version, protocol version, and publication semantics. Its date, scientific rules, and artifact-content rules were unchanged.

Each destination is staged at a temporary path in its own parent and installed with atomic no-replace semantics. On the official macOS/MPS runtime, the private files and temporary directory must pass the F_FULLFSYNC durability barrier. The private output directory is then installed and its parent flushed. The public summary is installed last as the completion marker, followed by a flush of its parent. Completion requires both destinations and verification of their installed identities.

Publication across the two destinations is not atomic. Abrupt process or host failure can leave private output without the public summary. Either one-sided state is classified as incomplete_not_result and left unchanged until an operator verifies that it is stale and removes it before a rerun. A run that finds both destinations already present is refused rather than replacing them. If an in-process BaseException is caught, cleanup removes only destinations created by that run. It attempts each removal independently and attempts the parent flushes without obscuring the original publication failure.

The implementation included the fitting procedure, tests, command-line interface and private/public artifact publication controls. Commit c3a5c815b20121f1ddd06a2f316f904077c00c4f was published on September 17, 2026, and its CI run passed: https://github.com/KrtiT/automated-phishing-detection-public/actions/runs/35257955477. The official MPS attempt began at 2026-09-17T18:21:12Z from a clean detached checkout of that commit, using only the pinned training and validation inputs. It stopped at 2026-09-17T19:24:27Z with exit code 2 after 3,795.28 seconds: "stage-one artifact threshold does not match the supplied scores." Neither output destination was present, and there are no accepted transformer or cascade artifacts from that attempt. The subsequent controlled retry and its accepted artifacts are reported in Section 4.2. CI verifies code checks, not scientific outcomes. Seeds 42 through 46 supplied the declared sensitivity analysis, and Random Forest supplied a secondary benchmark. Their results are reported in Section 4.5 and do not determine a primary hypothesis.

Protocol v1.10 governs the research record; its matrix SHA-256 is aad6b7cf8d8416bfb37ec19503dbed031c15767ea96d9b76486ca3f3fedebe1b. It freezes

the GMM development method as rq2-gmm-development-v1, SHA-256 22d32088b05e74432704f9671ab76ba28b4f573ead418846b23bc366315cb393. This prospective method freeze changes neither the RQ1 procedure nor any H1, H2 or H3 gate.

The preceding v1.9 matrix has SHA-256 f24eac919cb79d24d2248a94b3a74208f7b4d809ad778b963ad2e62315d78a38. Commits 7205e5630c5d7805d9bf6d293ead14b89cfe10a5 and 0d45625527040632d444e552f9a25ccf39a3fabf record the v2 publication amendment and clarification. The published, CI-verified first-attempt commit is c3a5c815b20121f1ddd06a2f316f904077c00c4f. The active transformer/cascade contract SHA-256 is 686c0d86b33b8a6c2e09cd6e174003db0bd2f7c30b087faf5470e6a270524213.

The September 3 advisor report SHA-256 is b72da89a4cc8a5b06f6ca88d79fe78dd54e3199a96b7450209ea53b4a4c04215. The aggregate baseline-validation record was added at commit 3078e39515513df398392be625a3ee21f1ef5b4b; its summary SHA-256 is bf5b3a6f0fc705d26852da4dd0053c6111ffc3e500d7a2e95dfba5ad859b279c. These code, contract and CI identities establish provenance, not predictive performance.

The controlled retry completed at 2026-09-17T22:19:28Z from commit e866441f2ff858472d031b8d358fd469897c6a65. Its accepted summary has SHA-256 41499aa388babe60442de7231b4087f67a53f96f340568a7cc58a3268606a2fd. The separate GMM development run completed with false_alert_gate_met=false. Those

development commands used only the pinned training and validation inputs; group-test and PhishVN records were excluded from those commands, then accessed under the separate study authorization described in Section 4.6. H1 and H3 are not supported by the complete results in Chapter 4. H2 is not supported under its original conjunction because a required component failed; favorable external results cannot reverse that conclusion. Historical summaries retain their original status fields rather than being rewritten.

The development runner verifies the accepted training and validation identities, reviewed code and fixed runtime. It reserves the run before reading inputs and records each comparison separately. Training references are constructed before validation access, and each completed comparison is saved before the next begins. A failed computation stops the run without retrying. Earlier outputs are retained, and the remaining comparisons are marked as unattempted. The parent process requires a successful worker exit and checks receipts, output hashes, model identities, and saved CP, AP, AUC and drift-calibration calculations. It neither refits a model nor reopens the source records. Tests use invented data; the research runs are reported separately.

The separate seed/probe runner uses one fresh worker for seed-42 calibration, each new seed, and the probe replay. It reserves each stage before reading its inputs. Consumed training identities and labels, epoch probabilities, qualifying checkpoints and restored state are retained before later checks; shared stage-one predictions are stored once. The verifier recomputes label digests, checkpoint selection, calibration and decisions from saved evidence. It checks token/mask descriptor structure, but does not re-encode those arrays without source URLs. Probe records distinguish scored rows from completed

routing and window results. A stopped process preserves its evidence and actual exit, while the remaining stages stay unattempted.

These procedures were tested on invented inputs, including fresh subprocesses, abrupt worker termination and altered evidence. The later v2 execution retained five completed seed-stage summaries but stopped before probe replay, as reported in Section 4.6. Those summaries were preliminary at that stop. Section 4.5 records their later all-or-none acceptance by a separate no-fit retained-stage audit and the separately authorized probe correction; that acceptance neither promotes the failed v2 root nor selects, refits or drops a seed. Saved-evidence verification is not an independent refit or primary-model rescoring of the original sources. The method choices were made after earlier development observations and are not represented as having preceded them.

The September 30 completion directive authorized bounded checkpoint implementation, retained-evidence verification and conditional continuation, not waiver of scientific or physical gates. The exact continuation code was frozen at 77d128377ce5b401437d7179f5cd78fb4294b72c with a hash-bound study-only profile and execution envelope. The sealed history selected only attempt four; attempts one through three supplied no substitute observations. Both scientific source worker exits, all accepted cells, preparation ancestry and stop history were checked before continuation. The final verification reauthenticated that evidence and recomputed saved-outcome counts, primary intervals and pooled latency reductions without fitting or executing a model. It observed all 250 owned service/client exits as successful. In the October 1 segment, 1,945 retained condition samples recorded AC, sleep inhibition, no thermal/performance warning and no detected known competing workload. Sampled

known-command checks cannot exclude every possible workload or establish uninterrupted conditions between samples. Root exit was zero with no recorded session violation. This procedural verification is not independent external replication or advisor approval.

Final timing requires observed AC power, normal available thermal/performance reports, owned sleep inhibition and a reserved host without concurrent tests, training, builds or benchmarks. Conditions are sampled before and after arms and approximately every five seconds during execution. This is sampled observation with operator reservation, not proof of continuously idle hardware. Successful, failed and all-request latency denominators, throughput, physical counters and process exits remain distinct. The new synthetic workload and altered client cannot replace the original H3 measurements or establish a transformer speedup.

A dated October 2 recovery amendment superseded only the initial no-additional-schedule rule after an observed power interruption. The first 45 complete arms and interrupted 46th arm were preserved; no primary structural c64 arm had begun. Following the operator's explicit approval, a separate hash-bound launcher reused the unchanged frozen source, all 80 arms, thresholds, deadlines and strict response semantics. It required at least 180 seconds of sampled stable AC and retained the original continuous guard loop during one new complete schedule. No partial arm or complete arm from the first attempt contributes to the primary S result. The amendment was made after partial nonprimary observations, not backdated, and asserted no advisor or institutional approval.

Final verification checked the saved evidence through 7,340 file-hash comparisons and 238,159 assertions. It authenticated both historical scientific worker exits and all 250 actual service/client exits. Separate read-only calculations recomputed the primary counts, six clustered contrasts and pooled terminal latency reductions from saved observations.

The continuation completed at 2026-10-01T22:05:09Z with root exit 0 and no recorded session violation. No model was refitted, no new predictions were made for synthesis and no original dataset was reopened. Here, independent recomputation refers to a separate calculation, not an independent investigator or replication.

The evidence record distinguishes accepted estimates from diagnostic and development runs. It retains the initial baseline nonconvergence and provenance-incomplete diagnostic, together with the platform scoring stop, transformer reconstruction mismatch, two secondary-development failures and seed/probe metadata stop. These records and their corrections appear in the history supplement. The corrections restored scoring and artifact-consistency checks without relaxing a scientific threshold. The accepted Random Forest correction refitted only its declared RF model; the seed/probe correction accepted all five retained stages without refitting and performed one authorized probe execution. The predecessor records remain separate from the accepted artifacts, preserving the distinction between correction of execution and evidence of model performance.

External input preparation identified a representation mismatch before prediction. Applying the raw-field parser to 8,941 external test rows retained 294 records: 266 certified and 28 bronze. The exclusions comprised 8,553 invalid-URL records and 94

overlaps, leaving no eligible gold or Tranco population and activating the whole-study hold. All 69 publisher gold values and all 1,241 publisher Tranco raw values lacked schemes. The subsequent input amendment adopted the publisher's exact url_norm consistently for parsing and model input, while retaining the raw cells and the original overlap, model, threshold and hypothesis rules. Its timing is material: the decision followed preparation and label-count inspection but preceded amended predictions. It was an operator-authorized methodological change, rather than a wholly pre-access specification or an assertion of advisor approval.

Operational completion used one explicitly selected evidence prefix. Four interrupted attempts preceded the accepted continuation: the first two contained seven and fifteen completed cells, the third stopped during source evaluation, and the fourth contained 72 completed cells. The checkpoint amendment was defined after source predictions and partial measurement. It selected attempt four only and required both source outputs and the entire 72-cell prefix to pass the eligibility review before the remaining 53 cells could be measured in their original order. All four attempt records and partial cells remain preserved. No result from attempts one through three replaced an accepted observation, no partial requests from stopped cell 73 entered its complete measurement, and no accepted adverse cell was repeated to obtain a better result.

The complete service comparison on October 2, 2026, contains 80 arms: two workloads, two concurrency levels, ten pairs and two client topologies. It contributes 800,000 measured requests and 80,000 warmup requests, with all 80 service processes exiting normally. The host was reserved for the comparison. Recorded condition checks comprised 37 preflight samples over 184.66 seconds and 414 schedule observations

covering external power, available thermal/performance status, sleep inhibition and competing workloads. These sampled checks document the measurement environment; they do not establish continuous absence of disturbance. The complete schedule is analyzed independently of the preceding interrupted schedule.

A preceding schedule ended at an environmental stop condition after 45 completed arms and part of the next arm. Its 450,000 completed measured requests had no recorded request errors, and no primary structural concurrency-64 arm had begun. A methodological amendment then specified one new full 80-arm schedule after stable preflight conditions. This decision followed partial nonprimary observations and is distinguished from the original pre-experiment specification. All 372 files from the earlier schedule, including the partial arm and stop record, remain hash-identical and excluded from the new primary result. The analysis neither pools the schedules nor selects favorable arms across them.

Independent arithmetic recomputation from saved JavaScript Object Notation (JSON) records verifies all 80 arm summaries, 40 paired comparisons, eight pooled groups and five S requirements against the frozen reducer. It performs no predictions or fitting and is not independent-investigator replication. The new schedule records one shared-client timeout and zero warmup errors. The original study's 125 cells, 25 groups, 22 primary checks and 1,901 recorded errors remain separate and unchanged.

The post-review P/F/R/C execution used frozen research code f77027b3963f4e7ee6b831b9f09ff04c71491001 and profile SHA-256 84e4d47f2c7c6249766dc677b6b7c2cad5be8fdaefb65195d786be418463c4d6. All four

tasks completed on October 12, 2026 (Pacific time). The separate saved-evidence audit reauthenticated bound inputs, exact consumed labels, saved predictions, routing decisions, measured request membership, quantiles and actual execution counters without fitting or predicting. Raw cohort identities and execution metadata remain private; the public revision contains their releasable aggregate evidence and methods.

Preserved introduction before final length review

1.1 Background

Networked applications, including artificial-intelligence (AI) services, receive and produce uniform resource locators (URLs) through application programming interfaces, retrieval pipelines, tool calls, and model-generated responses. A URL at this boundary may be legitimate, or it may direct a user or downstream system to a credential-harvesting site. An inference gateway therefore presents a useful point for a narrow control: inspect the raw URL before it continues through the application. This control does not replace browser, email, endpoint, or network defenses. It addresses the smaller question of whether a gateway can identify suspicious URLs with sufficiently low false-positive and operational costs to be useful inline.

Raw-URL detection is deliberately constrained. The detector in this study does not fetch the destination, render page content, query the Domain Name System (DNS) or the WHOIS registration-record service, or use private request content. It observes the URL string and features derived from that string. This boundary supports repeatable evaluation and avoids turning data collection into live interaction with suspected phishing

infrastructure. It also makes the scientific question harder: the system must distinguish records using syntax, character sequence, and structural patterns that may vary across domains and data sources.

Two practical tensions motivate the work. First, a random row split can place related domains in both training and test sets, allowing domain-family overlap to contribute to an apparently strong generalization result. Second, a character transformer may recognize patterns unavailable to a linear structural model, but its potential detection benefit must be weighed against inference cost. Selective execution offers one way to manage that tradeoff, while distribution monitoring can identify circumstances in which the original routing rule may warrant reconsideration. The research problem is whether these established components, combined in one gateway, satisfy the same detection and service requirements across the declared evaluation populations.

1.2 Problem Statement

Published phishing-URL studies often optimize aggregate accuracy or report performance under dataset-specific splits. An inference-gateway operator instead needs evidence at a low false-positive operating point, on domains not shared with model fitting, and on an external source whose collection process differs from the development source. The operator also needs to know how often a more expensive model is invoked and what the complete Hypertext Transfer Protocol (HTTP) path costs under load. These concerns are related but are not answered by one accuracy value.

Public-corpus labels require careful interpretation. A binary label records a reference classification supplied through the corpus's collection process; it is not necessarily a

current, independently adjudicated fact about a live site. Silently changing those labels, filling missing records or selecting thresholds after observing test outcomes would weaken reproducibility and risk overstating performance.

The protocol used staged freezes to separate development choices from evaluation: mechanical label mapping, quarantine, domain allocation, model and cascade threshold selection, and hypothesis decision rules each had an explicit boundary. Protocol v1.10 fixed the Gaussian mixture model (GMM) development method before its run. Two later methodological changes had different information boundaries. The external input specification adopted the publisher's url_norm after preparation and label-count inspection; the two-session completion rule followed partial execution. Sections 3.10, 3.13 and 4.6 distinguish these changes from the earlier freezes so that the reader can assess which decisions preceded which observations.

1.3 Purpose of the Study

This study evaluates CyberSentinel, a two-stage URL-only detector designed for an inference-gateway boundary. Stage 1 uses L1-regularized logistic regression over locally derived structural features; Stage 2 uses a compact character-level transformer. A validation-selected uncertainty band determines escalation in the fixed cascade. A Gaussian mixture model monitors the structural features and stage-1 score. Its routing policy is evaluated through offline, ordered-stream replay: an alert changes routing for the next 256 requests, never for the records that generated it. The term prospective routing denotes this future-only action rule, not prospective field collection.

The evaluation connects three questions: what richer representations contribute at a low false-positive-rate (FPR) operating point, whether distribution alerts improve escalation decisions, and what detection and service costs determine inline feasibility. PhiUSIIL supplies development and registrable-domain-disjoint internal evaluation; PhishVN v3.1.0, Mendeley Data repository Version 4, supplies external source/domain evaluation. The investigation completes both source evaluations, all 125 operational cells in 25 groups and all 22 primary checks. Each primary conclusion is tied to its population, metric and numerical requirement. The post-review analysis corrects paired uncertainty without changing those requirements. The secondary analyses examine ranking, calibration, comparator behavior and robustness; Section 4.5 distinguishes the source-identifiability analysis and permutation sensitivity results from validated negative controls. Eight primary component checks pass after the paired-inference correction in Section 3.8. Chapters 4 and 5 explain why their benefits do not satisfy all conditions of H1–H3, and then evaluate two targeted modifications to representation and service execution.

The initial findings motivated two subsequent, bounded comparisons. Detection comparison D evaluated a transport-neutral structural representation, implemented by consistent scheme neutralization, on an additional admitted benchmark. Service comparison S evaluated client connection ownership while retaining the structural scorer. Together, these comparisons extend the account from design and evaluation through diagnosis, modification and measured comparison. They establish an exact representation property and a substantial service-latency improvement, while quantifying the conditions

those improvements did not resolve. Their requirements were specified after the initial findings and remain distinct from H1–H3.

Preserved literature exposition before length review

2.2 Structural and Character Representations

A sparse linear model provides a useful first stage because its inputs and coefficient path are inspectable. L1 regularization can suppress weak or redundant predictors while retaining a small set of nonzero coefficients (Tibshirani, 1996). It does not, however, directly model long character sequences or interactions among distant parts of a URL. A character-level transformer can learn sequence patterns without tokenizing the URL as natural language. That flexibility may improve recall, but it increases fitting and inference cost and can also learn collection-source shortcuts.

Ahamed et al. (2026) evaluate structural and character-based URL approaches under adversarial, domain-disjoint, and external conditions. Their principal PhishTank external check combines external phishing positives with held-out PhiUSIIL benign records, while a separate dual-source check uses an independent top-sites negative source. Their work therefore supports the need for cross-source evaluation without making every reported external result equivalent to a wholly independent two-class corpus. It also narrows what may be claimed here: combining structural and character models is not by itself new. The present RQ1 instead asks whether each added representation provides a prespecified incremental recall gain at a controlled FPR on both an internal domain-disjoint partition and external source/domain strata.

The Random Forest comparator examines a different source of representational capacity from a character encoder. Breiman (2001) constructs ensembles of randomized tree predictors and relates forest behavior to the strength and correlation of their component trees. Trees can represent nonlinear combinations of supplied features without learning a character representation. In this study, that comparison asks whether a richer decision function over structural variables is enough; it does not make tree performance a proxy for sequence-model performance. The sparse linear, tree and character models therefore address different mechanisms rather than forming an assumed ranking from simple to best.

Character-level learning predates the transformer used here. Zhang et al. (2015) investigate character-level convolutional networks for text classification. Le et al. (2018) introduce URLNet, which jointly learns character- and word-level convolutional URL representations for malicious-URL detection. Neither model is the present transformer. Vaswani et al. (2017) establish attention-based sequence modeling in machine translation; applying that architecture family to URL characters is an application choice, not an invention of attention. These distinctions prevent an architectural label from replacing the empirical question: does the additional sequence information improve recall at the locked low-false-positive operating point on the relevant populations?

Pedregosa et al. (2011) describe scikit-learn as a consistent interface to supervised and unsupervised learning methods. That citation identifies the software used for classical-model development, but neither library use nor a passing software test establishes predictive validity. Chapter 3 therefore identifies the fitted parameters, feature definitions, package versions, partitions and frozen thresholds of the executed artifact.

These records make the implementation inspectable; they do not reproduce the library's original research evaluation.

2.3 Domain Leakage and Source Shift

Random row allocation is problematic when many URLs share a registrable domain. Related records in training and test data can make a model appear to generalize while it is partly recognizing domain-specific regularities. A registrable-domain group split is therefore used for PhiUSIIL. Each retained registrable domain belongs to one and only one partition, and the allocation is label-blind.

Domain separation does not eliminate source effects. A detector trained on one corpus may learn formatting, curation, or collection artifacts that do not transfer to another. Rashid et al. (2024) document cross-dataset degradation and study unsupervised domain adaptation for phishing URLs. Tsai et al. (2024) examine dataset bias in malicious-URL models and adversarial training for invariant representations. These findings support the need for an external source/domain evaluation, but they do not predetermine its outcome.

The external evaluation uses PhishVN v3.1.0 from Mendeley Data repository Version 4 under the staged freezes and disclosed representation amendment in Section 3.10. Vu (2026a) describes source records, confidence tiers, and registrable-domain-grouped splitting for that corpus. The current study does not treat all external records as equally strong outcome evidence. National Cyber Security Center (NCSC) gold positives and certified trusted-registry negatives form the primary external error strata. Tranco records are a popularity-based control rather than verified benign evidence. Le Pochat et al. (2019) show how the composition, stability and susceptibility to manipulation of

popularity rankings can affect samples used in security research. Their findings support keeping popularity-based controls separate from independently verified negative labels.

Mechanical validity and conflict rules are applied before routing to define the retained external stream. Among retained records, source and confidence designations are applied only after label-blind routing to form analytic outcome strata; they do not change routing membership.

The reason for grouping is dependence, not a presumption that every domain has one label. Roberts et al. (2017) examine blocked validation for temporally, spatially and hierarchically structured ecological data. Their application is not phishing, but the methodological warning is relevant: random allocation can underestimate predictive error when related observations cross the train–test boundary. They also show why blocking changes the prediction problem and can introduce extrapolation. A domain-disjoint score should therefore be interpreted as performance on the chosen held-out domain population, not as a universally more accurate estimate of every deployment setting. Grouping controls a named leakage route; it does not certify labels or remove all collection artifacts.

Pan and Yang (2010) distinguish transfer-learning settings and relate them to domain adaptation and sample-selection bias. Ben-David et al. (2010) bound target error using source error and a classifier-induced divergence under an assumption that a hypothesis can perform well in both domains. This establishes why small source error alone is insufficient and why a distribution comparison requires assumptions before it becomes a performance guarantee. The present study does not estimate that theoretical bound. Its

external strata instead test the behavior of the unchanged fitted detector on a specified source mixture.

Covariate shift is a narrower condition than arbitrary source change. Sugiyama et al. (2007) study importance-weighted cross-validation when the input distribution changes but the conditional output distribution given the input remains unchanged. A new phishing feed may violate that condition through changes in labeling, collection practices or the relationship between URL structure and reference class. Importance weighting therefore cannot be invoked as an automatic remedy without checking the assumptions and defining a new development procedure. This study implements neither importance-weighted model selection nor target-domain adaptation. Its source-separated results identify where a frozen model transfers and where further development would require a new evaluation, rather than silently turning the external test into training data.

2.4 Calibration at Low False-Positive Rates

An inline alerting system can be unusable even with high aggregate accuracy if its false-positive rate is too large. Axelsson's (2000) base-rate analysis of intrusion detection explains why rare events can make false alarms dominate the alert population despite high detection sensitivity. That analysis motivates examining low-FPR behavior, but does not prescribe the particular 1% ceiling adopted here. Threshold selection therefore belongs to the validation phase rather than the test phase. In this protocol, a model threshold maximizes validation recall only among thresholds whose exact one-sided 95% Clopper-Pearson FPR upper bound is <= 1% (Clopper & Pearson, 1934). Ties favor the smaller upper bound and then the higher threshold. If no threshold satisfies the constraint, the study records that outcome rather than weakening the rule.

This design distinguishes an observed rate from uncertainty about that rate. A point estimate of 1% based on a small negative sample does not carry the same evidentiary weight as the same estimate based on a large sample. Exact binomial bounds make the sample denominator visible. The 1% ceiling remains a study-defined risk choice rather than a universal threshold derived from earlier work.

Different metrics answer different operational questions. Saito and Rehmsmeier (2015) show why precision–recall views are important when class imbalance can obscure the practical meaning of receiver operating characteristic (ROC) plots. Precision depends on the proportion of positive examples in the evaluated population; a favorable value on a balanced benchmark cannot be transferred unchanged to a stream containing far fewer phishing requests. Conversely, a ROC area summarizes ranking across thresholds and does not establish the recall available at one low false-positive rate. Reporting both broad discrimination and the frozen operating point makes these interpretations distinguishable.

Chicco and Jurman (2020) explain how accuracy and the F1 score can conceal poor binary-classification behavior and discuss the Matthews correlation coefficient (MCC) as a summary that incorporates all four confusion-matrix categories. That motivates reporting complementary summaries and the underlying counts rather than declaring one scalar sufficient. MCC still does not encode this gateway's false-positive ceiling or service budget. The primary decision follows the specified constraint, even when another descriptive metric improves.

Probabilistic scores introduce another distinction. Gneiting and Raftery (2007) develop proper scoring rules for evaluating probabilistic forecasts, including quadratic and

logarithmic scores. Such rules assess the quality of probability assignments against observed outcomes; they do not by themselves select an operational action threshold. Calibration diagnostics, ranking metrics and confusion counts at the locked threshold are therefore complementary, not interchangeable. This distinction is particularly important when a representation modification produces more stable scores but a different recall–specificity tradeoff. Improved behavior under one diagnostic must not be restated as satisfaction of a separate low-FPR requirement.

Prevalence projections require a further distinction between changing class proportions and changing class-conditional behavior. Lipton et al. (2018) study label shift, in which P(Y) changes while P(X|Y) remains fixed, and develop a method for estimating the changed class proportions. The prevalence projections reported here hold the measured class-conditional rates fixed and vary an assumed phishing proportion; they neither estimate a deployment prevalence nor implement that paper's shift-correction method. Their value is to show how the alert burden depends on the assumed class mixture, separately from the measured source-transfer problem.

2.6 Distribution Monitoring and Routing

A Gaussian mixture model (GMM) represents the development feature distribution as a weighted combination of Gaussian components. In the expectation–maximization formulation, component membership is unobserved; estimation alternates between conditional membership weights and parameter updates (Dempster et al., 1977). The primary monitor uses standardized structural features plus the stage-1 score and selects one through six diagonal-covariance components by the Bayesian information criterion (Schwarz, 1978). Window scores are mean negative log-likelihood. The alert boundary is

calibrated on one PhiUSIIL validation stream, while false alerts are measured on a separate validation audit stream.

Security drift research shows the value of detecting and explaining distributional changes. CADE, for example, detects and explains individual drift samples through a learned low-dimensional representation in malware and network-intrusion settings (Yang et al., 2021). It does not validate the present diagonal-GMM likelihood, overlapping-window, alert-boundary, or next-256 routing rules. The inference supported by the present GMM is narrower. A high negative log-likelihood signals that observed X values differ from the fitted P(X) reference. The alert does not reveal why the distribution changed and does not prove that labels or prediction errors changed. Comparing fixed-cascade errors with future-only alert-policy errors is therefore necessary to test routing value.

Drift literature separates recognizing change from responding successfully to it. Gama et al. (2014) review adaptive learning under changes in the relation between inputs and targets. Lu et al. (2019) organize learning under concept drift around detection, understanding and adaptation. These distinctions are essential here because an unlabeled feature monitor cannot directly observe the conditional error process. A signal that inputs are unusual may justify investigation, but automatically routing future traffic to a more expensive model adds a separate intervention whose benefit must be measured. The fixed detector and future-only routing policy deliberately keep the observation and the action distinguishable.

Chandola et al. (2009) organize anomaly-detection methods by their assumptions about normal and anomalous behavior. A low density under a fitted reference distribution is

consequently model-relative, not a declaration that a URL is malicious. Common benign inputs may look unusual to a reference fitted on another source, while harmful inputs may resemble that reference. This is why the monitor has both a false-alert audit and a downstream error comparison. Counting detected external windows without those checks would answer only the easier distribution-recognition question.

Gretton et al. (2012) introduce maximum mean discrepancy (MMD) as a kernel statistic for comparing distributions. It provides a different reference for asking whether feature samples differ, whereas the Gaussian mixture monitor scores observations against an explicitly fitted density. The MMD comparison in this study is a distribution diagnostic; the kernel-test literature does not confer a phishing-error guarantee on it or on the mixture monitor. Overlapping windows and a future-only action horizon further require attention to the actual unit of observation. Window detection, request classification and additional model execution have different denominators, which are kept separate in the results.

2.8 Evaluation-Guided Engineering Iteration

An engineering evaluation has to distinguish a correct implementation from a useful operating point. Arp et al. (2022) describe how spurious correlations, inappropriate evaluation sets and test-informed development can produce misleading security-machine-learning conclusions. Their analysis motivates two decisions here. First, a diagnostic observation is used to formulate a mechanism to test, rather than to declare its cause established. Second, observations already used to diagnose a model are treated as development evidence for any revision. They do not become untouched evaluation data merely because a new model is fitted or a different split is selected.

The distinction between domain, source and temporal separation is equally important. TESSERACT formalizes spatial and temporal evaluation constraints for malware classification (Pendlebury et al., 2019). Its application differs from phishing URLs, but the methodological distinction transfers: excluding overlapping registrable domains does not establish forward-in-time performance. The additional URL benchmark in this study was collected in 2020, before the reported phishing collection period of PhiUSIIL. It therefore supports a retrospective cross-dataset comparison, not a claim of resilience against later attacks. Shared upstream phishing feeds also prevent treating different dataset titles as proof of independent collection mechanisms.

Calibration, discrimination and action are separate properties. Guo et al. (2017) calibrate confidence using validation data; their temperature-scaling method does not alter the predicted class ranking in the manner required to cure arbitrary discrimination failures. In a binary setting, a positive-temperature monotone transformation cannot improve ROC ranking. It would therefore be unjustified to infer that recalibration alone must repair poor low-FPR external behavior. Similarly, SelectiveNet studies risk subject to coverage constraints with an explicit reject option (Geifman & El-Yaniv, 2019). It supports reporting the decisions and coverage produced by selection, rather than counting deferred cases as corrected predictions. Neither temperature scaling nor SelectiveNet is introduced as an implemented algorithm in this study.

These principles organize a bounded engineering iteration. The initial evaluation identifies a representation sensitivity and a possible client-side cost mechanism. The follow-up tests one representation change and one connection-management change while retaining their respective comparators. Representation stability, labeled detection

performance and HTTP performance remain separate measured outcomes. The value is the traceable progression from an observed behavior to a constrained modification and a comparison capable of showing whether that modification addresses the intended requirement.

Metamorphic testing makes one part of that iteration more precise. Chen et al. (2018) describe relations between multiple inputs and their expected outputs as a means of testing systems when an ordinary output oracle is difficult to supply. The prescribed HTTP/HTTPS companion transformation here tests a declared representation relation: changing the transport scheme alone should not change the transport-neutral feature vector or its score. It does not assert that two live pages served over different schemes have identical content or security. No page is fetched to establish such equivalence. Exact invariance under the prescribed transformation is therefore a conformance result with its own value, while labeled detection performance remains a separate empirical test. Keeping these two claims apart allows the study to document a successful implementation repair without concealing the tradeoff observed against publisher reference labels.

2.9 Operational Measurement and Evidence Units

An inference endpoint includes more than a classifier. Connection management, request scheduling, validation and response processing can affect observed latency even when model parameters are unchanged. Dean and Barroso (2013) explain why latency variability matters for the responsiveness of large interactive services. Their scale differs from this local gateway, but the distinction supports examining the slow tail rather than average service time alone. Kalibera and Jones (2013) emphasize repetition and

uncertainty estimation in systems benchmarking because execution varies across runs and other experimental levels. Their work motivates identifying the level at which comparisons are replicated rather than treating a large request count as a large number of independent experiments. The present service comparison uses paired runs for the latency-ratio interval and separately reports pooled request-level quantiles. These summaries describe different quantities and must not be substituted for one another.

The workload generator also determines what a throughput or latency result means. Schroeder et al. (2006) distinguish closed workloads, in which completions govern subsequent submissions, from open workloads with arrivals independent of completion. Their experiments show that these choices can produce materially different response times and scheduling behavior. The present fixed-concurrency replays therefore characterize the tested closed-loop service path. They support a controlled comparison of client designs, rather than a capacity claim for an externally imposed production arrival rate.

Service correctness also has multiple layers. Agreement of predictive fields answers whether the two client paths return the same classification information on comparable requests. Exact response agreement tests the wider response contract, including the admission sequence retained by the frozen rule. Concurrent clients can preserve predictions while changing that sequence. A timeout creates an additional noncomparable pair rather than a correct prediction. The service experiment consequently reports the full attempted denominator, errors, exact agreement and prediction-only agreement. The literature supports careful measurement; the particular agreement rule and latency budget remain study-defined requirements, not thresholds borrowed from a benchmark paper.

Across these literatures, the contribution is the connection among explicit boundaries: what information the detector receives, which population it is asked to generalize to, how its operating point is selected, what the monitor can observe, what routing changes and what the service timer includes. The experiment exposes these boundaries through common artifacts and preserved decisions. That approach yields interpretable positive engineering findings and identifies the conditions under which they do not establish the larger deployment claim.